



News-Driven Liquidity Dynamics in West Texas Intermediate Crude Oil Futures

A Channel-Decomposition Approach

Enrique Favila Martinez
S1173176

Supervisor: Prof. Dr. Lejla Batina

Second reader: Prof. Dr. Tom Heskes

Erasmus Mundus Joint Master in Artificial Intelligence

Radboud Universiteit, Faculty of Science

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Glossary

This glossary defines terms from finance, statistics, and machine learning that recur in this thesis, in plain language, for readers who are not specialists in these fields. Terms are listed alphabetically.

Ablation study. Removing or changing one component of a model at a time to measure how much that component matters, like testing a recipe by leaving out one ingredient at a time.

Amihud illiquidity ratio. A measure of how much the price moves for a given amount of trading. A high value means even small trades push the price a lot, so the market is illiquid.

Asymmetry (directional). When two opposite cases produce different-sized effects. Here, whether negative news moves the market more than positive news of the same strength.

Bearish / Bullish. Bearish means expecting the price to fall (news that is bad for the price). Bullish means expecting it to rise (news that is good for the price).

Channel (supply, demand, risk premium). In this thesis, one of three separate economic dimensions a news item can affect: how much oil is available (supply), how much is consumed (demand), or how much geopolitical or operational risk it implies (risk premium).

Chokepoint. A narrow, critical passage in a supply route (for example the Strait of Hormuz) whose disruption can cut off a large share of the world's oil flow.

Coefficient. In a regression, a number that says how much the outcome changes when a given input increases by one unit.

Contemporaneous. Happening at the same time. A contemporaneous effect is felt in the same hour the news arrives, with no delay.

Cross-sectional. An analysis that treats each observation (here, each article) on its own, rather than as a connected sequence over time.

DXY (U.S. Dollar Index). A measure of the dollar's value against a basket of major currencies. Oil is priced in dollars, so it is a relevant control variable.

EIA (Energy Information Administration). The U.S. agency that publishes weekly crude oil inventory figures, a regularly scheduled and closely watched event in oil markets.

Embedding. A way of turning a category (such as “supply” versus “demand”) into a small list of numbers the model can learn from, instead of an arbitrary code.

Encoder / Decoder. In a forecasting model, the encoder reads the past window of data and the decoder produces the forecast for the future steps.

Estimand. The specific quantity a method is actually estimating. Two methods can both study “asymmetry” yet target different estimands, so their answers need not agree.

Front-month futures. The futures contract closest to expiry, which is the most actively traded one for a commodity like crude oil.

Futures contract. An agreement to buy or sell an asset at a set price on a future date. It is the main way crude oil is traded.

Lag. A delay. A “+6 hour lag” effect is one that appears six hours after the news, not immediately.

Liquidity. How easily an asset can be traded without moving its price. It is measured here mainly by hourly trading volume.

Logarithm (log), log volume. A mathematical transform that compresses large numbers. Taking the log of trading volume makes its large swings easier to model.

Marginal effect. The change in the outcome from a one-unit increase in one input, holding everything else fixed.

Multi-horizon forecast. A forecast that predicts several steps ahead at once (here, 1, 3, 6, and 12 hours), rather than only the next step.

OLS (ordinary least squares). The standard method for fitting a straight-line (linear) relationship between inputs and an outcome.

OPEC / OPEC+. The group of major oil-producing countries whose production decisions strongly affect global oil supply and price.

Orthogonal. Independent and non-overlapping. Orthogonal channels each capture a distinct dimension without duplicating one another.

Parkinson range. A volatility measure based on the gap between an hour’s highest and lowest prices.

Persistence baseline. A simple forecast that just repeats the last observed value (“the next value equals the current one”). Models are judged by how far they beat it.

p-value. A number indicating how likely a result could have appeared by chance. Small values (for example below 0.05) suggest the result is real rather than accidental.

Quantile. A cut-point in a range of possible values (the median is the 50% quantile). The model predicts a spread of quantiles to express its uncertainty.

Regime, structural break. A regime is a distinct market state (for example calm versus wartime). A structural break is the point where the market shifts from one regime to another.

Risk premium. The extra compensation markets demand for bearing risk. Here, how much a news item raises uncertainty about oil supply.

Salience. How attention-grabbing or prominent something is, regardless of whether it is good or bad.

Sentiment score. A numerical reading of how positive or negative a text is. In this thesis it specifically means the net effect the news implies for the WTI price.

Structured extraction. Having a model fill in a fixed set of fields (a schema) from free text, instead of returning a single label.

Temporal Fusion Transformer (TFT). A neural-network forecaster for time series designed to be interpretable, so one can read off which inputs and which past hours it relied on.

Tone versus price impact. Tone is how positive or negative the words of an article sound. Price impact is what the news actually implies for the price. The two can disagree: a supply threat sounds alarming (negative tone) yet pushes prices up (bullish).

VAR (vector autoregression). A time-series model of how several variables move together and influence one another over time.

Variable Selection Network (VSN). The part of the Temporal Fusion Transformer that learns how much weight to give each input feature.

VIX. A market index measuring the stock market’s expected volatility, often called the “fear gauge”.

Volatility. How much a price fluctuates. Higher volatility means larger, faster swings.

Volume (trading volume). The number of contracts traded in a period. It is the primary liquidity measure in this thesis.

WTI (West Texas Intermediate). The U.S. benchmark grade of crude oil, and the instrument this thesis studies.

1. Introduction

1.1 Motivation

News moves markets. This is not a new claim. [Tetlock \[2007\]](#) showed that pessimism in the financial press predicts downward pressure on prices, and that unusually high or low media sentiment predicts high trading volume. The problem is that most of this work is done on daily data. A daily bar treats the whole session as one number, and the interesting part happens inside it. When a scheduled release lands, the market reacts within minutes to hours and then the reaction fades over the rest of the day [[Ederington and Lee, 1993](#)]. Daily data cannot tell us whether liquidity moves on impact, an hour later, or only the next day. It has already averaged that away.

West Texas Intermediate (WTI) crude oil is a good place to look at this more closely. The news that drives it is frequent and, more importantly, well dated: Organization of the Petroleum Exporting Countries (OPEC) policy decisions, geopolitics in the producing regions, the weekly U.S. Energy Information Administration (EIA) inventory release, and the usual macroeconomic announcements. Each of these arrives at a time we know, and each could plausibly move liquidity over a few hours rather than a few days. WTI is also deeply liquid, so there is enough trading activity to measure at a fine grain. That combination is what makes an hourly study possible here.

1.2 Problem statement

The question this thesis works on is how news propagates into WTI liquidity at hourly resolution. Three things need answering.

The first is timing. When news arrives, does trading react in the same hour, or later? If later, how much later?

The second is direction. Does bearish news move liquidity more than bullish news of the same size? Loss aversion says it should, so it is worth testing.

The third is context. Liquidity does not respond to news in a vacuum. Some of the response belongs to the news, and some belongs to the state of the market at the time: how volatile things already were, where the dollar was, what kind of event it was. We want to know how that splits, and we want to read the split off a model rather than assume it.

Getting to any of this means two things. Unstructured news text has to become numbers on an hourly grid lined up with market data. Then the news-liquidity relationship has to be modelled with methods transparent enough that the timing and direction results can actually be interpreted, rather than just reported.

1.3 Research questions

RQ1 (lag structure). At what lag, if any, does news sentiment exert its strongest effect on WTI trading liquidity, measured at hourly resolution?

RQ2 (directional asymmetry). Does the liquidity response differ between bearish and bullish news, and if so, in what sense?

1.4 Contributions

There are three contributions defined in this thesis. Chapters 4 and 5 develop them deeper.

(a) The intraday shape of the news-liquidity response, and its directional asymmetry. Most news-and-markets work runs at daily resolution. That is enough to establish *that* news moves trading activity [Tetlock, 2007], but not *when* inside the session, because a daily bar has averaged the answer away before the analysis begins. The intraday work that does exist looks mainly at scheduled macroeconomic announcements, and at prices and volatility, where most of the adjustment completes within minutes of release [Ederington and Lee, 1993]. This thesis measures a different object: how *liquidity* responds to a continuous stream of largely unscheduled news, hour by hour.

Working at that resolution is what makes three results visible, none of which a daily design can represent:

- **The response is not contemporaneous.** It peaks at +6 hours and has decayed to insignificance by +8, so the entire lifetime of the effect fits inside a single daily bar (Section 4.2.3).
- **The decay is not smooth.** Secondary maxima at +1h and +4h point to discrete information-incorporation events rather than a continuous fade.
- **The timing itself is direction-dependent.** Bearish news is integrated over roughly six hours against one hour for bullish, an asymmetry that lives in *when* the market responds rather than in how much (Section 4.3.7.5).

Two methods with no assumptions in common agree on the +6h peak: a lag-regression baseline (Section 4.2.3) and the attention pattern of a Temporal Fusion Transformer (Section 4.3.7). Read against Ederington and Lee [1993], where prices finish adjusting

within minutes, the substantive claim is that **liquidity is the slower variable**: the market has repriced long before it has finished trading on the news. Reaching that result also meant solving an alignment problem specific to sub-daily work, since news publishes continuously while trading does not, so every article has to be placed on a grid with overnight and weekend gaps in it (Section 3.2).

Asymmetry is more involved. The two phases do not measure the same quantity, so their results look like a disagreement at first and are not one. The linear baseline gives a larger marginal coefficient on negative sentiment than on positive (Section 4.2.3). The forecasting model finds no bearish-versus-bullish gap in predicted volume at all, and instead leans on what the news implies about risk: market volatility and the extracted supply channel are its two highest-weighted features, while the composite sentiment score ranks 21st of 90 (Section 4.3.7). Put together, the conclusion is not that bearish news beats bullish news. It is that liquidity cares more about how much risk the news carries than about which way it points.

(b) A channel decomposition of LLM-extracted news features, checked by cross-model agreement. Collapsing an article to one sentiment score throws information away. Phase 2 instead extracts separate supply, demand, and risk-premium channels with a large language model, following the structural decomposition of oil-price shocks in Kilian [2009]. There is no human ground truth to check the extraction against, so we check it a different way: two models are run over the same articles, and we ask how much they agree. Recalibrating onto the channels lifts cross-model correlation on the composite sentiment from 0.39 to 0.88 (Section 4.3.4). The decomposition also clears up something that looks odd otherwise, which is why supply-risk geopolitical news comes out price-bullish rather than bearish.

(c) A comparison of title-only and full-body news representations. Sentiment taken from headlines alone disagrees with sentiment from full article bodies on 41.6% of articles. The disagreement is not random either: when an article flips, it flips bearish far more often than bullish (Section 4.2.2). Studies that use titles alone are picking up this bias whether they intend to or not, and this puts a number on it.

There is a common thread. Phase 1 is a simple, interpretable baseline (FinBERT features, lag regression). Phase 2 is richer and more expressive (LLM channels, a Temporal Fusion Transformer). The second corroborates the first and gives us more to say about it, and neither one gets thrown away (Section 5.3).

1.5 Structure of the thesis

On Chapter 2: Background, we remind the terminology and places this thesis uses against prior work on news and markets, financial-text models, and interpretable forecasting. On Chapter 3: Methods, we then cover the two-phase methodology: getting the data and aligning it to an hourly grid, extracting features by phases (FinBERT in Phase 1, LLM channel extraction in Phase 2), and the modelling and evaluation setup. On Chapter 4: Experiments we report the experiment, what the headline-bias comparison means and the lag regression from Phase 1, following the Temporal Fusion Transformer results from Phase 2 with its feature importance, temporal attention, and the directional-asymmetry test. Later on Chapter 5: Discussion, reads the findings back against the research questions and sets out the limitations found in the experimentation phase. Finally, Chapter 6: Conclusion closes the thesis giving out what we learnt and what we could work on in the future. After the main chapters, the appendices hold the extraction prompt and schema, the ablation and hyperparameters, the canonical entity list, and a reproducibility statement mentioned through the previous chapters.

2. Background

This chapter covers what is needed to follow the methods (Chapter 3) and the experiments (Chapter 4): the WTI market and how we measure its liquidity, the NLP tools that turn news into features, the Temporal Fusion Transformer, and the prior work behind the design choices made here.

2.1 Terminology

This section fixes the vocabulary. Each idea is picked up again where it is actually used, in Chapters 3 and 4.

WTI crude oil futures and the EIA inventory cycle. West Texas Intermediate (WTI) is the U.S. benchmark grade of crude oil. This thesis works on its front-month futures contract, which is the most actively traded WTI instrument. We picked WTI for three reasons: it is deeply liquid, it reacts to an event structure we can pin down (geopolitics, OPEC policy, macroeconomic releases), and it has a weekly clock. That clock is the U.S. Energy Information Administration (EIA) Weekly Petroleum Status Report, which publishes national crude inventory levels every Wednesday at 10:30 a.m. Eastern Time. It is the most regular scheduled event in the oil calendar.

Liquidity measures. Liquidity is how easily an asset can be traded without pushing its price around. One number will not capture that, so we use three, all computed hourly. Trading volume, in log form, is the raw activity. The Amihud illiquidity ratio [Amihud, 2002] is the absolute return per unit of volume, which tells us about price impact: how far the price moves for a given amount of trading. The Parkinson range [Parkinson, 1980] is the log high-low range within the hour, and it estimates volatility at high frequency. Between them we get activity, price impact, and volatility.

NLP for finance. Turning news text into numbers uses three ideas from natural language processing. Three-class sentiment classification labels a text positive, neutral, or negative, and it is what financial sentiment models normally output. Structured extraction asks the model to fill in a fixed schema instead, say a sentiment score, an event type, and a list of entities, which gives features with more than one dimension. Prompt engineering is how a large language model is told to produce that structured output, and tool-use schema enforcement is the stricter version of it, where the output shape is imposed rather than

requested.

Temporal Fusion Transformer. The Temporal Fusion Transformer (TFT) of [Lim et al. \[2021\]](#) is an attention-based neural network for multi-horizon time-series forecasting. Two parts of it matter for this thesis because they are what make it interpretable. The Variable Selection Network gives every input feature a learned importance weight. The interpretable multi-head attention layer shows how much the model looks at each past time step when it forecasts. Between them, we can read what the model is doing instead of treating it as a black box.

2.2 Related work

The design decisions here come from prior work rather than from first principles. This section lays out the anchors, and each one comes back at the point where it is used in Chapters 3 and 4.

News sentiment and market activity. The starting premise is that news sentiment moves trading activity and not just prices, and the finance literature already supports it. [Tetlock \[2007\]](#) found that media pessimism predicts downward price pressure, while unusually high or low media sentiment predicts high trading volume. That is what justifies using news sentiment to predict hourly liquidity in Phase 1 (Chapter 4, Section 4.2).

News at intraday resolution. The reaction to news is not spread evenly across the day, it is bunched up. [Ederington and Lee \[1993\]](#) showed that scheduled macroeconomic announcements account for most of the intraday and day-of-week volatility patterns in interest-rate and foreign-exchange futures, with most of the adjustment happening within minutes of release and volatility staying high for several hours after. Despite this, most news-sentiment work in finance still runs at daily resolution, which averages away the exact structure we are trying to measure. Working hourly (Chapter 3) is the main methodological choice that follows from this.

News in oil markets. At coarse resolution, oil barely reacts to news. [Kilian and Vega \[2011\]](#) found no compelling response of daily energy-price changes to daily U.S. macroeconomic news releases, which fits the idea that energy prices are predetermined with respect to macro aggregates at a daily horizon. A null at daily frequency does not rule out an effect at finer resolution. That is part of why this thesis goes hourly, and why the target is liquidity (trading volume) rather than price.

Financial-text sentiment models. Transformer models adapted to the domain beat general-purpose sentiment tools on financial language. FinBERT [\[Araci, 2019\]](#) is the de facto baseline for financial sentiment, and we use it as the Phase 1 sentiment extractor (Section 3.3.1). Its output is only three classes, though, and that coarseness is exactly

what motivates the richer LLM extraction in Phase 2.

Liquidity measurement. Hourly liquidity is measured with standard market-microstructure quantities: trading volume, the Parkinson high-low range [Parkinson, 1980], and the Amihud illiquidity ratio [Amihud, 2002], which captures price impact per unit of traded volume.

Oil-price shock decomposition. An oil-market move is not one undifferentiated shock. It is a mix of drivers that can be separated, and the canonical way of separating them is the structural decomposition of the real oil price into supply, aggregate-demand, and precautionary (risk) demand components [Kilian, 2009]. This is the basis for the Phase 2 channel decomposition (Sections 3.3.4 and 4.3.2), which pulls a `supply_impact`, a `demand_impact`, and a `risk_premium` out of each article rather than squashing them onto one sentiment axis. It also explains why supply-risk geopolitical news is price-bullish rather than bearish.

Interpretable multi-horizon forecasting. The Temporal Fusion Transformer [Lim et al., 2021] forecasts over multiple horizons while staying interpretable, through its Variable Selection Network and its interpretable multi-head attention. Those two mechanisms are what let Phase 2 (Sections 3.6, 4.3.5, and 4.3.7) read feature importance and temporal attention straight off the model. We use them directly to corroborate the lag structure in RQ1.

LLM-based extraction and cross-model validation. Large language models can pull structured, multi-dimensional features out of financial text. They can also stand in for annotation work that would otherwise be expensive: Gilardi et al. [2023] found they match or beat crowd workers on text labelling, and Zheng et al. [2023] used them to evaluate other models’ outputs. Phase 2 leans on both. Claude Haiku extracts the schema in Section 3.3.2, giving the richer, price-oriented, channel-decomposed features that FinBERT’s three-class tone label cannot produce. The inter-model calibration of Section 3.7 (Section 4.3.4) then validates those features by checking them against a second model.

3. Methods

3.1 Data sources and acquisition

Three data sources go into one hourly dataset for WTI crude oil: market data on prices and trading activity, structured macroeconomic indicators, and unstructured news. Each one arrives on its own clock and is accessed in its own way, so getting them onto a shared time grid is a real methodological problem rather than a formatting step. Section 3.2 deals with it.

3.1.1 Commodity selection

We settled on WTI crude oil futures after screening a few candidates. Sugar futures were the first alternative considered and were dropped: the reason for this is that prices moved too little and reacted too weakly to news at hourly resolution, and the research questions here need both. WTI has the opposite profile. Its event structure is rich, driven by geopolitics, OPEC policy, the weekly EIA inventory cycle, and macroeconomic forces, and those events show up as hourly volume and price reactions we can actually measure.

3.1.2 Market data

Hourly Open, Highest, Lowest, Close, Volume (OHLCV) data are pulled with the yfinance Python library [Aroussi, 2024] for three instruments:

- **CL=F**: WTI crude oil front-month futures contract, the primary target series
- **DX-Y.NYB**: the U.S. Dollar Index (DXY), used as a macroeconomic covariate
- **^VIX**: the Chicago Board Options Exchange (CBOE) Volatility Index (VIX), used as a market-stress covariate

DXY and VIX come in as exogenous controls in the Phase 2 pipeline (Section 4.3), which lets us separate the news signal from wider macro-financial movement. They are not used in the Phase 1 analyses (Section 4.2), which look at news sentiment against trading volume directly, without macro controls.

From the raw OHLCV records we derive three liquidity and volatility proxies plus one return measure, all hourly:

- **log_volume**: the natural logarithm of contracts traded per hour, used as the primary liquidity measure
- **price_range**: the hourly Parkinson range [Parkinson, 1980], computed as $\log(\text{high}) - \log(\text{low})$, a high-frequency volatility proxy
- **log_return**: the hourly log return, $\log(\text{close}_t / \text{close}_{t-1})$
- **amihud**: the Amihud illiquidity ratio at hourly resolution, computed as $|\log_return| / \text{volume}$ following Amihud [2002]. The ratio is undefined for hours with zero volume, so those rows are dropped from any analysis where Amihud is the dependent variable.

The canonical market grid runs from 13 May 2024 to 13 May 2026, which gives 11,232 hourly observations over roughly 24 months of trading. Phase 1 (Section 4.2.1) ran on an earlier pull that reached further back, roughly March 2024 to March 2026, for 11,219 aligned market hours, and we keep those values with the Phase 1 features rather than recomputing them. The reason the two windows differ is simple: yfinance only serves hourly data for a rolling window of about the last two years. By the time we re-pulled the market data and extended the corpus through the post-war period for Phase 2 (Section 4.3.1), the earliest hours of 2024 had aged out of that window, so the Phase 2 grid starts in May 2024. The gap between the two counts comes from the re-pull and nothing else. No result depends on it and the methodology did not change.

3.1.3 EIA inventory data

Weekly U.S. commercial crude oil inventory reports come from the U.S. Energy Information Administration (EIA) v2 API [U.S. Energy Information Administration, 2024]. We query the WCRSTUS1 series (weekly ending stocks of crude oil and petroleum products, U.S. total) with `frequency=weekly`. We download the full history from January 2020 onward so that rolling baselines can be computed, even though only the 2024–2026 window is modelled.

Each weekly release becomes two derived features, available hourly once temporal alignment has been applied:

- **is_eia_release**: a binary indicator that fires on the trading hour containing or immediately following the official release time
- **eia_surprise**: the weekly inventory change relative to a rolling four-week baseline, capturing the directional surprise component of each release

EIA releases land on Wednesdays at 10:30 a.m. U.S. Eastern Time, and they are a scheduled event that the oil market watches closely, which is why we built features from them in

the first place. In the reported models, though, the EIA cycle only enters through the `is_wednesday` calendar proxy (Section 3.6.2). We computed and stored `is_eia_release` and `eia_surprise` and tested them in an extra model run, but the continuous surprise carried almost no feature importance and the release flag added little on top of the calendar proxy, so neither made it into the canonical model.

3.1.4 News data

News comes from the GDELT Project [Leetaru and Schrod, 2013], a large machine-monitored corpus covering broadcast, print, and web sources worldwide. We use the GDELT 2.0 Document API to pull article metadata (URL, source, publication time, title) across a set of thematic queries aimed at WTI-relevant content.

The query set changed between the two phases. Phase 1 stuck to direct oil-market terms, giving eight queries:

```
crude oil market
OPEC production cut
oil supply demand
petroleum inventory
Brent crude price
oil inventory report
energy market outlook
crude oil WTI price
```

The Phase 1 scrape returned 51,948 raw article records over the March 2024 to February 2026 window. Deduplication and English-only filtering cut that to 16,326 unique articles. To get body content we HTTP-fetched each URL and parsed the page. The failures clustered on paywalled domains, Cloudflare-protected sources, and pages that need JavaScript to render. An allow-list heuristic regex function (Section 3.4) then judged 7,755 of the retrieved bodies to be substantive. Phase 1 analyses ran on a combined set of 13,690 articles: 7,755 with valid bodies going into title-plus-body sentiment extraction, plus 5,935 where the body fetch failed and we kept the title alone as a fallback.

For Phase 2 we widened the query set to catch the geopolitical and demand-side themes that dominated the post-war coverage period (Section 4.3.1):

```
Iran sanctions oil
Saudi Arabia oil production
China oil demand
Russia oil exports
oil supply disruption
```

Re-running the scrape over the full window through May 2026 with both query sets returned 76,345 raw article records spanning January 2024 to May 2026. Deduplication and English-only filtering left 22,795 unique articles, which is the modeling-ready Phase 2 corpus (Section 4.3.1). Of those, 19,619 produced a real body. The other 3,176 came back as a null response, a Cloudflare or paywall block, or an explicit fetch error. Phase 2 pushes articles without bodies into LLM-based filtering instead of falling back to titles the way Phase 1 did, and Section 3.4 explains why.

3.1.5 Persistence

The main tables are `articles` (raw GDELT records and scraped bodies), `market_context` (the hourly grid of derived WTI liquidity features with DXY and VIX levels), `eia_events` (raw EIA releases), `llm_features` (LLM-extracted news features, Phase 2 only), `article_entities` (canonical entity mentions per article, which is what the entity flags are built from), and `liquidity` (the modeling-ready joined view, populated by the alignment step in Section 3.2). The original Phase 1 pipeline passed CSV intermediates between notebooks, which let state drift out of sync. Putting everything in one relational database fixed that and made the canonical alignment a single deterministic join.

3.2 Temporal alignment

The three sources run on different clocks. Market data sits on a clean grid of trading hours, each bar tagged with its closing timestamp. EIA releases are weekly events at a fixed time. News articles are the awkward one: they can be published at any timestamp at all, including overnight, at weekends, and on holidays when nobody is trading. If we want a model that conditions liquidity at hour t on news published before t , every article needs to be pinned to a specific hour on the market grid.

We handle this with two rules, both applied when articles are written to the database: ceiling assignment to the next trading hour, and forward-assignment of off-hours articles to the next available trading hour.

3.2.1 Ceiling assignment to the next trading hour

For an article published at timestamp τ , the assigned market hour is

$$\text{datetime_hour}(\tau) = \lceil \tau \rceil_{1h}. \quad (3.1)$$

So an article published at 14:23 UTC goes to the 15:00 bar, not the 14:00 one. The reason is causal. The 14:00 bar covers 14:00 to 15:00, so part of its volume has already formed by 14:23. Assigning the article to the 14:00 bar would let news influence a candle that

started forming before the news existed, which is backwards. The ceiling rule makes sure the article strictly precedes the start of the hour it is matched to.

The ceiling rule is conservative. It pushes news to the next full hour even when the article might well have moved the market within minutes, so it may understate the immediate intra-hour reaction. We accept that, because the alternative bias is worse: attributing pre-publication price movement to post-publication news would be a direct threat to the causal reading. The bias runs in the safe direction, understating rather than inflating the immediate reaction, and we come back to this where the lag results are interpreted (Section 4.2.3).

3.2.2 Forward-assignment of off-hours articles

A lot of the corpus is published when no trading is happening: overnight, at weekends, or during holiday closures. There were two options. Drop those articles, or push them forward to the next available trading hour. We went with the second.

The reasoning is empirical. Geopolitical events, supply shocks, and weekend OPEC communiqués get covered during off-hours as a matter of routine, and they are exactly the events that produce the biggest reactions once trading restarts. Dropping them would cut the most informative part of the corpus, and it would do so systematically rather than at random. Forward-assignment keeps that information. The price is a measurement-time lag between when an article was published and the hour it gets assigned to.

In practice, when an article’s ceiling-assigned hour is not on the market grid because the market is shut, we assign it to the next available `market_context` row in chronological order. This is a vectorised pandas `merge_asof` join with `direction='forward'`, which does the whole corpus in one pass.

We keep a diagnostic column, `assignment_gap`, which is the difference in hours between the assigned market hour and the original publication timestamp:

$$\text{assignment_gap} = \text{datetime_hour} - \text{publication_timestamp}. \quad (3.2)$$

For an article published during a session this is at most one hour. For weekend or overnight articles it can run to many hours, or across a multi-day holiday closure in the worst case. The gap lives in the `liquidity` table and gets used downstream in two ways: as a control variable, and as the basis for stratified analyses separating *contemporaneous* news ($\text{gap} < 2\text{h}$) from *forward-assigned* news ($\text{gap} \geq 2\text{h}$). The exploratory Phase 1 VAR (later abandoned, Section 4.2.4) only used contemporaneous articles aggregated to hourly resolution. The Phase 2 TFT models use the whole corpus and let the model decide whether contemporaneous and forward-assigned articles deserve different weight.

3.2.3 Joined output

Alignment produces the `liquidity` table, where each row is one article carrying the market-context features for its assigned hour. One boundary effect is worth flagging. Under forward-assignment (Section 3.2.2), articles published before the market grid starts get mapped to the first available trading hour instead of being dropped. In the Phase 2 corpus that dumps roughly 3,600 pre-coverage articles (published before May 2024) onto the first grid hour, so the models see one unusually high-count hour right at the start of the series. Excluding pre-coverage articles outright gets rid of the artifact, and that adjustment is applied in the v2.3 ablation.

The resulting `liquidity` table has 22,795 rows: 15,290 contemporaneous ($\text{gap} < 2\text{h}$) and 7,505 forward-assigned ($\text{gap} \geq 2\text{h}$). Some analyses track the two separately, but the primary modeling combines them.

3.3 News feature extraction approaches

News articles are text, and the models downstream need numbers. Getting from one to the other is the main NLP problem in this thesis. We do it two different ways across the two phases: FinBERT sentiment classification in Phase 1, and structured LLM extraction with an LLM in Phase 2. They differ in what they output, and also in what they assume a “news feature” is in the first place.

3.3.1 FinBERT sentiment extraction

Phase 1 uses FinBERT [Araci, 2019], a BERT variant fine-tuned on financial text for three-class sentiment (positive, neutral, negative). It is one of the most widely used sentiment models in financial NLP, which makes it a natural baseline.

We run it through the HuggingFace Transformers library [Wolf et al., 2020], on Apple Silicon GPU acceleration via the MPS backend. For each article the model returns softmax probabilities over the three classes. The predicted label is the argmax, and we record a scalar `confidence` as the largest class probability:

$$\text{confidence} = \max(P_{\text{positive}}, P_{\text{negative}}, P_{\text{neutral}}). \quad (3.3)$$

We keep the categorical label, all three class probabilities, and the confidence value for downstream use, persisted in the Phase 1 feature table `gdelt_wti_sentiment.csv`.

FinBERT runs twice per article:

- **Title-only input:** the article title fed alone to the model

- **Title-plus-body input:** the title concatenated with the body, subject to FinBERT’s 512-token input limit

Where the body has to be truncated, the title stays at the front of the sequence so the most salient part is never the thing that gets cut. Articles whose body retrieval failed (Section 3.1.4) fall back to title-only sentiment as their canonical input, and the body-availability status stays in the database so downstream analyses can tell the two regimes apart.

For the lag OLS analysis (Sections 3.5.1 and 4.2) we use the FinBERT class probabilities directly rather than the argmax label, so each article is weighted by how confident the model was instead of being collapsed to a hard vote. The lag OLS takes $P(\text{negative})$ and $P(\text{positive})$ as two separate continuous regressors, which is what keeps the bearish-versus-bullish asymmetry that RQ2 needs. The earlier discrete mapping $\{\text{positive} : +1, \text{neutral} : 0, \text{negative} : -1\}$ throws the probability magnitudes away, and we keep it only as a comparison baseline in Section 4.2.3. The title-only probabilities are held back for the comparison in Section 4.2.2, the headline bias finding, where they support both a categorical and a continuous divergence measure.

3.3.2 Structured LLM extraction

Phase 2 swaps FinBERT for Claude Haiku 4.5 [Anthropic, 2025a], a smaller model in Anthropic’s Claude family, and uses it to pull a richer set of structured features out of each article. In Chapter 4 (Section 4.3.2) we give the full argument for the move. The short version is that FinBERT’s three classes flatten something that is really continuous and multidimensional, and its confidence value (just the max class probability) does not carry much signal downstream.

The Haiku extraction went through one schema iteration. The first version (Haiku v1) produced the features for the first TFT training (Section 4.3.5) on the Phase 1 corpus. Per article it returned a continuous `sentiment_score` in $[-1, +1]$ for the article’s net directional impact on the WTI price (-1 strongly bearish, $+1$ strongly bullish, judged from the content rather than the headline tone), a `magnitude` score for how important the event is, a `price_direction` label, an `event_type` label, a list of `entities`, a `certainty` score, and a `time_horizon` label. The second version (Haiku v2) came out of an inter-model calibration finding documented in Section 4.3.4. It dropped `price_direction`, turned `event_type` from a single label into an array of one to three labels ordered by salience, added an explicit `usable` boolean for filtering (Section 3.4), and added three orthogonal economic channel scores (`supply_impact`, `demand_impact`, `risk_premium`), which Section 3.3.4 describes. Appendix A has the full Haiku v2 prompt and tool schema.

3.3.3 Tool-use API and schema enforcement

The Haiku extraction goes through Anthropic’s tool-use API [Anthropic, 2025b], with `tool_choice`. Tool-use enforces the schema at the API level: enumerated values for categorical fields, type constraints on numeric fields, and required-field validation. An earlier pass that just asked the model in the prompt to produce valid JSON gave us schema violations constantly, for example returning `slightly_bullish` where a numeric score belonged, or a `time_horizon` label that was not on the list. Tool-use enforcement validates the numeric ranges and the scalar enumerated fields at the API boundary and gets rid of almost all of that. A small residue survives wherever the schema is permissive: the free-text `event_type` array picked up an off-list label on 78 of 10,514 articles (0.74%, `structural`), and one article has a malformed `time_horizon` string. We treat both as noise downstream. So the enforcement removes the bulk of the parsing brittleness, not every last case.

The tool-use design also buys a short-circuit optimisation. When the model decides an article is unusable, whether that is a paywall block, a cookie notice, or off-topic content, only `usable` has to be set and every other field is optional. The model takes advantage of this and returns roughly 10 output tokens instead of 200 on unusable articles, which cuts the output-token cost of an unusable-article extraction by around 90%.

3.3.4 Channel decomposition

Beyond the composite sentiment value, the Haiku schema has three continuous scores: `supply_impact`, `demand_impact`, and `risk_premium`, each in $[-1, +1]$. The point of the three channels is to break the economic content of an article into dimensions that do not overlap:

- `supply_impact` measures the article’s implication for the quantity of oil available to the market. Positive values indicate supply increases (sanctions lifted, OPEC adding barrels, new pipeline online); negative values indicate supply decreases (refinery outages, OPEC cuts, infrastructure attacks).
- `demand_impact` measures the article’s implication for oil consumption. Positive values indicate demand strength (Fed easing, China stimulus, growth surprise); negative values indicate demand weakness (recession signals, Fed hawkishness, lockdowns).
- `risk_premium` measures the change in geopolitical or operational risk priced into oil. Positive values indicate escalation (military strikes on oil infrastructure, sanctions imposed); negative values indicate de-escalation (ceasefire, sanctions lifted).

The channels were not in the original Phase 2 schema. They were a response to a calibration failure on the composite `sentiment_score`, where Haiku and GPT-5 disagreed about

the sign of sentiment on high-magnitude geopolitical events. Chapter 4 documents the full reasoning: the calibration failure is in Section 4.3.4, and the decomposition with its re-calibration is in Section 4.3.6. Methodologically the channels do two jobs here. They are higher-resolution features for TFT v2 (Section 4.3.7), and they give us something defensible to put in place of human-rater agreement when validating LLM-extracted features (Section 4.3.4).

We keep the composite `sentiment_score` alongside the channels rather than replacing it. Three reasons. It keeps continuity with the Phase 1 headline bias experiment, which needs a single sentiment value. It lets us run ablations comparing the composite signal against the decomposed channels in the TFT (Section 4.3.7). And it hedges against the channels turning out to be individually noisy once run at full scale.

3.3.5 Magnitude, event type, entities, certainty, and time horizon

The rest of the Haiku schema covers dimensions of news content that do not overlap with each other. `magnitude` is a continuous score in $[0, 1]$ for how market-moving an event is, with anchors given in the prompt (0.0 = negligible event, 1.0 = historic; most articles land in the 0.1–0.4 range, with a major OPEC cut anchored at 0.9 and a full-scale geopolitical crisis at 1.0). `event_type` is an array of one to three labels from a fixed set of six (`geopolitical`, `supply`, `demand`, `macro`, `technical`, `other`), ordered by salience within the article. `entities` lists the named actors central to the article (countries, organisations, oil companies, key officials). `certainty` is a continuous score in $[0, 1]$ separating rumour, analyst forecast, and confirmed fact. `time_horizon` is a label from `immediate` (hours to a day), `short_term` (days to weeks), and `structural` (months or longer themes such as energy transition).

Every one of these encodes something FinBERT’s three classes cannot, and each one enters TFT v2 either as a continuous input or as a categorical that the Variable Selection Network can weight on its own. The raw `entities` lists get normalised in post-processing onto a fixed canonical set of 71 actors using an alias mapping, so “Tehran” and “Iranian” both map to `Iran`. At hourly aggregation each hour gets a binary vector of length 71 marking which canonical entities the dominant article of that hour named, and that vector enters TFT v2 as 71 entity-flag features. Appendix D has the full list. Section 4.3.2 discusses the rationale for each field, including why `event_type` is a list rather than a single value and why entity normalisation happens in post-processing instead of in the prompt.

3.4 Filtering strategies

A good fraction of what GDELT returns is not substantive news. The usual offenders are paywall placeholders, cookie consent notices, Cloudflare blocks, scraping error pages, generic stock-pick listicles, and articles that matched a query keyword but have nothing to do with oil markets. These have to be filtered out. Left in, they would inject noise into the sentiment signal and bias the models downstream.

Each phase filters differently. Phase 1 used a regex/keyword heuristic on the article body. Phase 2 replaced it with an LLM-judged binary call on whether the content is usable, exposed as the `usable` field in the Haiku extraction. Both survive as columns in the database (`body_valid` and `usable`), which lets us compare them directly in Section 4.3.3.

3.4.1 Regex/keyword heuristic (Phase 1)

The Phase 1 filter, `body_valid`, is an allow-list run over the article body. A body is valid only if it meets all four criteria:

- Minimum length of 400 characters (excluding whitespace and HTML residues)
- Presence of at least one term from a domain-specific keyword list covering energy, financial, and geopolitical vocabulary
- Absence of press-release boilerplate (vendor blacklist for known PR-distribution patterns)
- An all-caps ratio below 30%, computed over words of more than two characters in texts containing more than ten words (excluding scraped pages that are predominantly menu links, category headers, or stock-ticker listings)

The filter grew over the early part of the project. It started as a deny-list of known bad phrases (Cloudflare error text, cookie notices) and turned into the allow-list above. On the Phase 1 corpus of 16,326 articles with English text and a successful body fetch, it passed 7,755 articles (47.5%). The rest were either kept with title-only sentiment as a fallback or dropped from the analysis, depending on the notebook. Section 4.2.1 spells out which analyses used which subset.

The regex filter is fast, reproducible, and transparent, in the sense that anyone can read the rules and predict what they will do to a given article. Inspection also made its limits obvious. Some articles passed while being mostly boilerplate, long disclaimers or repeated company descriptions that happened to mention energy keywords. Others failed while carrying legitimate analyst commentary, either because the body came in just under the 400-character threshold or because the vocabulary did not line up closely enough with the

energy keyword list.

That is what pushed us to a better filter in Phase 2.

3.4.2 LLM-judged usable flag (Phase 2)

The Phase 2 filter is the `usable` boolean in the Haiku schema, which asks the LLM itself whether the body is substantive WTI-relevant content. The prompt tells it to return `usable=false` for anything clearly not about oil markets, macro, or oil-relevant geopolitics. When `usable=false`, the model is allowed and told to skip every other field, which is the short-circuit from Section 3.3.3.

Two things motivated moving from `body_valid` to `usable`. First, the LLM can read semantic content that a regex cannot match, so it can tell a real analyst piece from a press release that happens to be full of energy keywords. Second, the LLM’s judgement runs on the same content model as the rest of the feature extraction, so the filter and the features agree with each other: an article marked usable will produce features that are themselves usable.

Then, `usable` becomes the primary content filter in Phase 2. There is also a stricter variant, `usable_strict`, which additionally requires at least one of the three economic channels (`supply_impact`, `demand_impact`, `risk_premium`) to be non-zero. That drops 1,161 articles which the model judged topical but gave no channel signal at all. The point of the stricter filter is not that those articles are mistakes, and Section 4.3.3 looks at what they actually are. It is that the channels are the Phase 2 feature contribution, so an article carrying no channel signal has nothing to contribute to them. `usable_strict=1` is the canonical filter for TFT v2 training (Section 4.3.3), and `usable=1` is what the filter comparison uses. `body_valid` stays in the database for the comparison in Section 4.3.3 but is not applied as a filter in any modelling step.

3.4.3 Comparative analysis

Because both filter columns exist for the Phase 2 corpus, we can look at where they disagree. Of the 19,619 Phase 2 articles the LLM processed, 13,550 are flagged `body_valid=1` and 11,675 are flagged `usable=1`. They disagree on a non-trivial share of articles in both directions: articles the regex rejected that the LLM accepted as substantive (the regex’s false positives), and articles the regex accepted that the LLM rejected as boilerplate (its false negatives). Section 4.3.3 reports the full agreement analysis, with Cohen’s kappa and worked examples of the disagreements.

3.5 Statistical models

The empirical work on news impact rests on lag-structured ordinary least squares (OLS) regressions, set out in Section 3.5.1. These are the canonical statistical evidence for both the lag question (RQ1) and the bearish-versus-bullish asymmetry question (RQ2). We also fitted an exploratory vector autoregression during development and abandoned it. The sentiment signal is sparse and event-driven, with more than half of all hours carrying no contemporaneous news at all, and the VAR could not identify significant dynamics on it. We do not develop it as a method here. Section 4.2.4 notes what happened and how it pushed us toward the deep-learning model of Section 3.6.

3.5.1 Lag OLS specification

The lag OLS asks whether news sentiment published at hour t predicts trading volume at hour $t + k$, for k over a fixed set of lag horizons. At lag k the specification is

$$\log_volume_{t+k} = \beta_0 + \beta_1 P(\text{negative})_t + \beta_2 P(\text{positive})_t + \varepsilon_t, \quad (3.4)$$

where $P(\text{negative})_t$ and $P(\text{positive})_t$ are the FinBERT title-plus-body class probabilities for the article published at hour t . The neutral probability is the omitted reference, since the three probabilities sum to one and including all three would be perfectly collinear. So β_1 and β_2 give the expected difference in `log_volume` at hour $t + k$ for a maximally confident bearish or bullish article (class probability of one) against a neutral one.

Using two class probabilities instead of one signed sentiment regressor is not a detail, it is what makes the asymmetry question answerable. A signed specification forces the bearish coefficient to be exactly the negative of the bullish one, which assumes a symmetric response before any estimation happens. The two-probability version lets them move independently, and that is what RQ2 is asking about. This continuous encoding is the primary Phase 1 specification. It generalises the earlier two-dummy encoding, which has the same two-coefficient structure but with hard 0/1 indicators, by weighting each article by FinBERT’s confidence, so a 0.95-bearish article counts for more than a 0.51-bearish one. We keep the dummy version as a baseline in Section 4.2.3.

We estimate a separate regression at each lag $k \in \{1, 2, 3, 4, 6, 8, 12\}$. The set is dense at short horizons, where we expect information propagation to be detectable, and thinner at long ones, where we expect the effect to fade. At each lag we report the two coefficients, their standard errors, p-values, and the regression R^2 . The R^2 will be small, and that is fine: news sentiment is one of many things driving hourly trading volume, and the goal is to find a structured pattern of coefficients across lags, not to explain as much variance as possible. What we treat as evidence of a real lag effect is a coefficient significant at the

conventional $p < 0.05$ threshold, repeated across several adjacent lags, with a magnitude pattern that hangs together.

The contemporaneous case ($k = 0$) is reported separately as a baseline. At hour t the article has only just been published, so an effect there means the news moved the market inside the same one-hour bar. It turns out to be small but statistically significant (Section 4.2.3). That is not a null result: it says sentiment carries detectable signal from the moment of publication, which is what makes it worth tracing across the later lags. Section 4.2.3 reports it next to the lag regressions.

3.6 Temporal Fusion Transformer

The Temporal Fusion Transformer (TFT) [Lim et al., 2021], from Lim, Arık, Loeff, and Pfister, is a deep-learning architecture for multi-horizon time-series forecasting with a mix of continuous and categorical inputs. It is our primary deep-learning model for Phase 2. We train two instances, TFT v1 and TFT v2, which share an architectural skeleton but differ in their features and their training data. This section covers the architecture. The feature configurations are in Sections 4.3.5 and 4.3.7.

3.6.1 Architecture overview

The TFT puts four components together into one end-to-end model:

- A **Variable Selection Network (VSN)** at each input time step, which assigns a learned scalar weight to each input feature, allowing the model to suppress uninformative features automatically
- A pair of **LSTM encoder–decoder layers** that process the history (encoder) and the forecast horizon (decoder) at fixed resolution
- An **interpretable multi-head self-attention** layer over the encoded sequence, exposing which historical time steps the model attends to when producing each forecast
- A **quantile output head** that produces multiple quantile predictions in a single forward pass, enabling prediction intervals without separate models

Three of these give us something interpretability-wise. The Variable Selection Network produces per-feature importance scores that can be compared across features, which answers the question of which inputs the model is actually using. The attention layer produces per-time-step weights that lay out the temporal structure of the model’s reasoning, and that goes straight to RQ1 on the lag structure. The quantile output gives us uncertainty estimates without training separate models for them.

3.6.2 Input typing

The TFT sorts inputs into four categories:

- **Time-varying known reals:** features whose values are known at every time step in both the history and the forecast horizon. Ours are calendar features (`hour`, `day_of_week`, `month`) and binary regime indicators (`is_us_session`, `is_wednesday` as a proxy for the EIA release day).
- **Time-varying unknown reals:** features known only over the history, which have to be predicted for the forecast horizon. These are the target itself (`log_volume`), the market-context features (`price_range`, `log_return`, `amihud`, `dxy`, `vix`), the continuous Haiku features (the composite `sentiment_score`, the three channels, `magnitude`, `certainty`), and, in TFT v2, the 71 binary entity flags. Sections 4.3.5 and 4.3.7 give the per-instance feature lists.
- **Time-varying unknown categoricals:** categorical news features known only over the history. In TFT v2 these are the dominant article’s `event_type` and `time_horizon`, passed as learned embeddings rather than integer codes. Whether they are encoded as proper categoricals or as integers is a design axis, and the v2 ablation isolates its effect (Section 4.3.7.4).
- **Static categoricals:** features fixed for the whole series. With a single asset this comes down to one asset identifier (`asset = 'WTI'`).

The Variable Selection Network learns separate gating weights per category, so a feature that is informative in the history as an unknown real but useless as a calendar input gets weighted correctly in both contexts.

3.6.3 Forecast configuration

Both instances condition each forecast on a fixed 48-hour encoder window, which is two full daily cycles of context. The length is deliberate. The Phase 1 lag OLS (Section 4.2.3) put the news-impact peak at lag +6 hours, and 48 hours comfortably covers that peak plus the second-day “memory” effects we saw in earlier TFT runs.

Target and horizon differ between the instances. TFT v1 (Section 4.3.5) predicts one target, `log_volume`, one hour ahead. TFT v2 (Section 4.3.7) predicts three targets jointly, `log_volume`, `amihud`, and `price_range`, at four horizons (1, 3, 6, and 12 hours), so its prediction length is 12 hours and it emits a whole trajectory rather than one step. The +6 hour horizon is there to line up with the lag OLS peak.

Both are trained with a quantile loss, which gives point predictions at several quantiles in one forward pass and handles target outliers without needing distributional assumptions.

TFT v1 uses a single `QuantileLoss` on its one target. TFT v2 uses a `MultiLoss` summing a `QuantileLoss` over each of the three targets. The median quantile is the point forecast throughout.

3.6.4 Architectural hyperparameters

The TFT has a small set of hyperparameters controlling capacity. We hold the core width parameters constant across both instances, so that any difference between them comes from features and forecast configuration rather than from raw capacity:

- `hidden_size = 32`: the internal representation dimension
- `attention_head_size = 4`: the number of parallel attention heads
- `hidden_continuous_size = 16`: the embedding dimension for continuous variable processing
- `learning_rate = 1e-3`: the initial Adam optimizer learning rate, with on-plateau reduction (patience 3)

Two things do differ. `dropout` goes from 0.1 in v1 to 0.15 in v2, a small bump in regularization for v2’s bigger input space. And even with the width unchanged, the trainable-parameter count goes from roughly 113,000 in v1 to 298,329 in v2, because v2 brings in the 71 entity-flag inputs, two more targets, and the multi-horizon output head. Keeping the width fixed while the feature set and forecast configuration grow is the point: it keeps the instances comparable on architectural capacity, so what we are measuring is the effect of the Phase 2 feature work.

3.6.5 Training procedure

Training runs in PyTorch Lightning [Falcon and The PyTorch Lightning team, 2024] through the `pytorch-forecasting` library [Beitner, 2024]. Both instances use the same optimizer and regularization scaffolding: Adam at the learning rate above, gradient clipping at 0.1 to keep updates stable over long sequences, early stopping on validation loss with a minimum improvement of `1e-4`, and a Google Colab T4 GPU. The `pytorch-forecasting TimeSeriesDataSet` wrapper builds the fixed-length sliding-window samples, encodes categoricals consistently, constructs the input tensors, and applies group normalization. Held-out sets go through the same wrapper with `stop_randomization=True` so evaluation is deterministic.

Three training settings differ, reflecting the difference in scale:

- **Temporal split.** TFT v1 uses an 80/20 split, the earlier 80% of hours for training and the most recent 20% for validation, with no held-out test set. TFT v2 uses a

60/20/20 split into train, validation, and test, with 48-hour buffers between partitions so encoder windows cannot leak across boundaries. The test set is what makes the pre-war versus war-regime evaluation of Section 4.3.7.3 possible.

- **Early stopping and epoch cap.** TFT v1 uses a patience of 5 epochs and a maximum of 30. TFT v2 raises these to 10 and 75. In practice early stopping halts both well before the cap.
- **Target normalization.** TFT v1 normalizes its single target with a `GroupNormalizer`. TFT v2 is multi-target, so it wraps one `GroupNormalizer` per target in a `MultiNormalizer`, scaling each of the three targets independently before the per-target losses are combined.

For reproducibility, the reported TFT v2 run also fixes the random seed (42) and uses deterministic algorithms.

3.6.6 Evaluation and interpretation outputs

Every trained TFT gives us a set of evaluation and interpretability outputs, all reported in Chapter 4.

Performance against a persistence baseline. The headline number is not the raw validation loss. The two instances train with different loss functions on different splits and horizons, so their loss values cannot be compared directly (Section 4.3.8). Instead we score forecasts by mean absolute error (MAE) against a persistence baseline, the naive forecast that carries the current value forward to the target horizon ($\hat{y}_{t+k} = y_t$), which is a standard reference for financial time series. What we report is the percentage reduction in MAE over persistence, which normalizes for how hard each target and horizon is. For TFT v2 we compute these for every target (`log_volume`, `amihud`, `price_range`) and horizon (1, 3, 6, 12 hours) on four slices: the validation set, the full test set, and the pre-war and war portions of the test set, split at the Iran war onset (Section 4.3.1). The pre-war and war slices are how we measure whether a model trained pre-war generalizes across the regime boundary.

Interpretability. Two outputs are read straight off each model, both averaged over the evaluation set: the Variable Selection Network feature-importance ranking, which shows which inputs the model leaned on, and the multi-head attention weights, which give a per-lag map of which historical hours it attended to.

Research-question diagnostics. The attention output matters most for RQ1. If TFT attention peaks at a lag consistent with the Phase 1 lag OLS result, then two analyses that start from completely different places agree: the OLS finds a structured lag effect from a series of single-lag linear specifications, while the TFT learns one nonlinear function

over the whole history and exposes the same structure through attention. For RQ2 we run a directional asymmetry test on the forecasts. Hours are split into bearish and bullish groups by the sign of their sentiment score, and a Welch t-test compares mean predicted `log_volume` between the groups at each horizon and slice. This is deliberately a different estimand from the lag OLS coefficient comparison: it contrasts the average level of predicted volume across two groups of hours, where the OLS measures the marginal sensitivity of volume to sentiment. Section 4.3.7.6 reads the two together with that distinction kept in view.

3.7 Inter-model calibration methodology

Everything in Phase 2 is built on the LLM-extracted features of Section 3.3. Before leaning on them we want an external check that the extraction is sensible and that the LLM is not systematically wrong on the cases that matter most. This section is the methodology for that check. The results, and the schema change they prompted, are in Chapter 4 (Sections 4.3.4 and 4.3.5).

3.7.1 The validation problem and our approach

Normally you validate an NLP extraction against human-annotated ground truth on a labelled subset. That was not available here. Annotating WTI news properly takes domain expertise in oil markets and trading that we don't have, and one non-expert annotator working at scale would add noise of unknown shape to the validation set rather than remove it. Hiring an expert at scale was beyond the project's resources.

So we use inter-model calibration instead, leaning on the evidence that LLMs are competent annotators in place of human labels [Gilardi et al., 2023] and can judge other models' output [Zheng et al., 2023]. The same articles, with the same prompt and tool schema, are scored by Claude Haiku 4.5 and by OpenAI's GPT-5. Disagreement between two models from different developer families is not the same signal as disagreement between humans, but it is still a signal. The argument goes as follows.

Each model carries its own implicit theory of how to read news, shaped by its training data, its alignment procedure, and the inductive biases of its developer. These two share little in either training data composition or alignment style. When they agree, that agreement is about content robust to two independently trained readings of financial news. When they disagree, it means either the article is genuinely ambiguous or one of the models is systematically biased.

This is weaker than expert-annotated ground truth and we are not claiming otherwise: two models agreeing on a wrong reading would sail straight through the comparison. Chapter 5

takes the limitation up as part of the discussion. It is still stronger than no external validation at all, and picking models from different families gets us as much diversity as possible given that both annotators are LLMs.

3.7.2 Sample design

The calibration sample is drawn by stratified random sampling from the full Phase 2 corpus. The strata come from the Phase 1 regex body-validity classification, in three buckets:

- **valid**: articles passing the regex `body_valid=1` filter (Section 3.4.1)
- **invalid_substantial**: articles failing `body_valid=1` but whose body field still contains substantial text content (length above a minimum threshold)
- **invalid_weak**: articles failing `body_valid=1` with bodies short enough to be likely scraping artefacts

We sample ten articles uniformly at random from each stratum, for a calibration sample of thirty. Stratifying this way makes sure the calibration sees the full range of content the LLM meets in production, including the marginal cases where the regex filter and the LLM are most likely to part ways.

3.7.3 Extraction procedure

Both models get the same system prompt and tool schema (Appendix A) for each of the thirty articles. The two output sets are then lined up by article identifier and compared field by field. The agreement metrics are computed symmetrically, so no metric depends on which model we call the reference.

3.7.4 Agreement metrics

For each field we compute one or more agreement measures, picked to suit the field's type:

- For binary fields (**usable**): agreement rate as a count out of thirty, and Cohen's kappa to correct for chance agreement
- For continuous fields (**sentiment_score**, **supply_impact**, **demand_impact**, **risk_premium**, **magnitude**, **certainty**): Pearson correlation across all pairs, mean absolute difference (MAD), and the count of **sign disagreements**, meaning cases where the two models' scores have opposite signs (one positive, one negative). Near-zero scores within a small range around zero count as agreement on neutrality and do not add to the sign-disagreement count.

- For categorical fields (`event_type`, `time_horizon`): exact-match agreement rate, and for the array-valued `event_type`, the proportion of cases where the top-ranked category from one model appears anywhere in the other model’s list

Of these, the sign-disagreement count matters most. A small mean absolute difference can hide cases where the two models disagree about the qualitative direction of an effect, one calling an event bullish and the other bearish, and that is exactly the disagreement that would flow into downstream models as noise. The sign-disagreement count puts those cases on the surface.

3.7.5 Use of the calibration outcome

The calibration sample is small ($n = 30$) on purpose. What we want is to spot qualitative patterns of agreement and disagreement, not to estimate a population agreement rate precisely. Thirty articles are enough to surface systematic disagreements, show what kind they are, and tell us whether to accept the schema or revise it.

If the calibration turns up significant disagreement on a field, the response is to work out where the disagreement comes from, revise the prompt or schema, then re-run the calibration on the same thirty articles with the new setup. Chapter 4 documents this loop: Section 4.3.4 has the first calibration result, and Section 4.3.6 has the schema revision and the re-run. Reusing the same thirty articles for both rounds is intentional, since it separates the effect of the schema change from the effect of sampling different articles.

4. Experiments

4.1 Overview

This chapter reports what both phases actually found. Phase 1 (Section 4.2) covers the regression work on the pre-war corpus: the headline bias experiment, the contemporaneous and lag OLS regressions that give the first evidence for RQ1 and RQ2, and the VAR. Phase 2 (Section 4.3) covers the deep-learning work on the expanded corpus: the LLM extraction pipeline, the two Temporal Fusion Transformers, and the ablation that shaped the v2 design. The chapter ends by comparing the two phases as methodologies (Section 4.3.8), setting out what Phase 2 changed and why putting the two phases’ numbers side by side would be a mistake.

4.2 Phase 1: initial pipeline and baseline modeling

Phase 1 builds the empirical baseline, starting from the existing evidence that news sentiment predicts trading activity and volume rather than just prices [Tetlock, 2007]. It runs on the FinBERT sentiment of Section 3.3.1 and the regex body-validity filter of Section 3.4.1, over WTI-relevant news published between March 2024 and February 2026. Three techniques are applied: a headline-bias comparison showing that title-only and title-plus-body sentiment disagree, linear regression (contemporaneous and lagged), and an exploratory vector autoregression that was abandoned. The headline-bias result is what removes the title-only fallback in Phase 2. By the end of Phase 1 we have the canonical statistical evidence for RQ1 (lag structure) and RQ2 (bearish-versus-bullish asymmetry), and the limits of a FinBERT-plus-classical-models pipeline are clear enough to justify the Phase 2 changes.

4.2.1 Phase 1 data and feature setup

The Phase 1 modeling dataset is **13,690 articles**: 7,755 with substantive bodies (regex filter `body_valid=1`, Section 3.4.1) scored on title plus body, and 5,935 scored on the title alone because no usable body was available. Those 5,935 split two ways: 1,881 where no body was retrieved at all, and 4,054 where a body was retrieved but did not pass the filter. Behind that are 51,948 raw GDELT records across eight oil-market queries, cut to 16,326 unique English articles after deduplication (Table 4.1). FinBERT [Araci, 2019] scores

each article (Section 3.3.1); the regressions use the title-plus-body class probabilities as the continuous sentiment feature, with the discrete $\{+1, 0, -1\}$ label kept as a baseline. For the 5,935 fallback articles that field holds the title-only probabilities, so the pooled feature mixes the two input regimes. Section 4.2.2 measures how far apart they are; because the disagreement runs one way, it attenuates rather than inflates the coefficients of Section 4.2.3. The sentiment splits **44.9% bearish**, **30.4% bullish**, **24.7% neutral**. A note on vocabulary before the results. FinBERT classifies the surface *tone* of financial text, not the direction of the WTI price; it has no asset grounding. “Bearish” and “bullish” throughout Phase 1 are therefore shorthand for its negative and positive tone labels. The two are not interchangeable: Section 4.3.8 shows that on geopolitical news they come apart, because a supply threat reads as negative in tone yet is bullish for price. The bearish skew is a property of headline tone, documented in Section 4.2.2, and should not be read as a claim about the population.

Articles get aligned to trading hours by the ceiling rule (Section 3.2.1), which puts 82.7% within two hours of a trading hour (contemporaneous) and forward-assigns the rest. The market backbone is 11,219 hourly WTI records carrying the liquidity targets (`log_volume`, `price_range`, `amihud` [Amihud, 2002]) from Section 3.1. DXY, VIX, and the EIA covariates only show up in Phase 2. Table 4.1 summarises how the dataset was built.

Table 4.1: Phase 1 dataset construction.

Stage	Count
Raw GDELT articles	51,948
After deduplication + English filter	16,326
With successful body retrieval	~13,000 (80% of 16,326)
With <code>body_valid=1</code> (regex-passed body)	7,755
Title-only fallback (no usable body; title retained)	5,935
Phase 1 modeling dataset	13,690
Aligned market hours (raw, before VAR cleaning)	11,219
Articles aligned within 2h gap (contemporaneous)	11,316
Articles aligned with gap ≥ 2 h (forward-assigned)	2,374
Article date range	2024-03-11 to 2026-03-01

Everything in the three Phase 1 experiments runs on this dataset.

4.2.2 Headline bias experiment

Motivation. The Phase 1 pipeline mixes two FinBERT input regimes in one 13,690-article dataset: title-only for the fallback articles, and title-plus-body for the rest. Before treating all of that as a single signal, we should check whether the two regimes actually agree. The

question is also interesting on its own. Headlines exist to get attention, and a body can qualify or even reverse whatever the title implies.

Method and categorical result. We took the 7,755 articles that have both a title and a substantive body and ran FinBERT twice on each, once on the title alone and once on title-plus-body. Because the model is the same both times, any difference comes from the input content (Section 3.3.1). On the argmax labels the two **disagree on 41.6% of articles** ($\chi^2 = 2050$, $p < 0.001$), and the disagreement has structure to it: positive titles keep a positive body only 59.2% of the time, with 40.8% moving toward neutral or negative; neutral titles resolve negative more often than positive (39.6% vs 30.0%); negative titles stay negative in 78.2% of cases.

Continuous result. The label measure has a blind spot. It ignores how far sentiment moves, and it cannot see any shift that stays inside one label. So we also define a signed score $P(\text{positive}) - P(\text{negative})$ per input and a divergence magnitude $|\text{signed_full} - \text{signed_title}|$. Label flips carry a mean magnitude of **0.96**, which is close to a full reversal. Same-label articles still move by **0.17** on average, with 31.6% moving more than 0.20. The mean signed shift from title to body is -0.09 , and 57.6% of articles get more bearish once the body is read.

Worked example. These quantities are easier to follow on real articles. The signed score is $P(\text{positive}) - P(\text{negative})$, computed once on the title and once on title plus body. The magnitude is the absolute distance between them, so how far sentiment moved, and the signed shift is $\text{signed_full} - \text{signed_title}$, so which way it moved.

Table 4.2: Worked examples of title-versus-body sentiment divergence.

Article (title)	signed_title	signed_full	magnitude	signed shift	Label transition
“Oil price soars above \$80”	+0.49	−0.63	1.12	−1.12	positive → negative (flip)
“U.S. Oil Is Stealing Market Share from OPEC+”	−0.32	−0.77	0.45	−0.45	negative → negative
“Oil prices continue to drop in world markets”	−0.96	−0.96	0.00	0.00	negative → negative

The first article flips. FinBERT reads the headline as bullish (“soars above \$80”) and then turns bearish once it sees the body, a near-full reversal at magnitude 1.12. The second keeps its negative label but still moves 0.45 further bearish, which is exactly the within-bucket movement the 41.6% flip rate cannot count. The third does not move at all, title and body agree. In all three the body is at least as bearish as the title, which is the one-directional pattern that produces the -0.09 mean signed shift.

Divergence and label-flip totals. Table 4.3 collects the aggregate quantities across the

7,755 title-plus-body articles.

Table 4.3: Title-versus-body divergence totals (7,755 articles).

Quantity	Value
Label agreement (title and body same label)	4,527 (58.4%)
Label flip (disagreement)	3,228 (41.6%)
Flip rate, positive titles	40.8%
Flip rate, neutral titles	69.5%
Flip rate, negative titles	21.8%
Mean divergence magnitude, all articles	0.50
Mean divergence magnitude, label flips	0.96
Mean divergence magnitude, same-label	0.17
Same-label articles with magnitude above 0.20	31.6%
Mean signed shift, title to body	−0.09
Articles more bearish once the body is read	57.6%

Implications. Two things carry forward. The first is that the input regime is not a detail: a pipeline that swaps title-only and title-plus-body freely is averaging over a 41.6% disagreement rate. The second cuts against the usual intuition. People expect attention-grabbing headlines to overstate negativity, but here **titles lean more bullish than the articles under them**, and reading the body pulls FinBERT’s sentiment more negative on average. Only the signed continuous score establishes that direction; the labels alone cannot. This matters directly for the regressions, because 5,935 of the 13,690 modelling rows carry a title-only score (Section 4.2.1). If those articles behave like the ones we can measure, their recorded sentiment sits above the value a usable body would have produced, so the sentiment regressor is measured with error on 43% of the sample. That caveat is worth stating plainly: the 41.6% figure comes from articles whose bodies passed the filter, and the fallback group need not behave identically. Measurement error in a regressor pulls estimated coefficients toward zero, so the effects reported in Section 4.2.3 are more likely conservative than inflated. Beyond Phase 1, this is the first concrete evidence that the choice of news representation matters, and it is what motivates the Phase 2 move to structured extraction from full bodies (Section 3.3.2).

4.2.3 Lag OLS: RQ1 and RQ2

The lag OLS regressions are the canonical Phase 1 evidence for RQ1 (the lag structure of news impact on liquidity) and RQ2 (bearish-versus-bullish asymmetry). They are also the result that the Phase 2 methodology is ultimately trying to corroborate.

Specification. At horizon k ,

$$\log_volume_{t+k} = \beta_0 + \beta_1 P(\text{negative})_t + \beta_2 P(\text{positive})_t + \varepsilon_t, \quad (4.1)$$

where $P(\text{negative})_t$ and $P(\text{positive})_t$ are the title-plus-body FinBERT class probabilities for the article published at hour t , and the neutral probability is the omitted reference (Section 3.5.1). So β_1 and β_2 give the `log_volume` response to a maximally confident bearish or bullish article against a neutral one. Each article is one observation, and we estimate a separate regression at each $k \in \{0, 1, 2, 3, 4, 6, 8, 12\}$, with $k = 0$ the contemporaneous case. Sample size runs from 13,690 rows at $k = 0$ down to roughly 9,900 at the longest lag. The continuous encoding is primary. The discrete-dummy baseline is kept for comparison, and at $k = 0$ it gives smaller coefficients ($\beta_{\text{bearish}} = 0.133$, $\beta_{\text{bullish}} = 0.103$) than the continuous 0.186 / 0.166, which confirms that weighting each article by FinBERT’s confidence sharpens the signal.

Table 4.4: Lag OLS coefficients across horizons.

Lag k (h)	$\beta_{P(\text{neg})}$	p	$\beta_{P(\text{pos})}$	p	R^2	n
0 (contemp.)	0.186	0.000	0.166	0.000	0.0015	13,690
1	0.262	0.000	0.200	0.000	0.0028	11,598
2	0.105	0.068	0.114	0.084	0.0003	11,512
3	0.117	0.047	0.170	0.013	0.0006	11,360
4	0.267	0.000	0.237	0.001	0.0017	11,065
6	0.342	0.000	0.291	0.000	0.0027	10,679
8	0.056	0.409	0.101	0.192	0.0002	10,440
12	0.001	0.984	-0.171	0.029	0.0008	9,917

Result. Table 4.4 has the regression at every horizon, one row per lag k in hours, with $k = 0$ the contemporaneous case. Reading the columns: $\beta_{P(\text{neg})}$ and $\beta_{P(\text{pos})}$ are the fitted coefficients on the bearish and bullish class probabilities, meaning the expected change in `log_volume` for a maximally confident bearish or bullish article against a neutral one. The two p columns are their two-sided p-values, with the usual significance threshold at $p < 0.05$. R^2 is the share of `log_volume` variance the two-feature regression explains at that lag. And n is how many articles enter that regression, which shrinks as k grows because targets at hour $t + k$ past the coverage window get dropped.

RQ1 (lag structure). The impact is detectable straight away at publication ($k = 0$), builds over the first few hours, peaks sharply at **+6h** ($\beta_{P(\text{neg})} = 0.342$, $\beta_{P(\text{pos})} = 0.291$, both $p < 0.001$), and has decayed to insignificance by +8h. To put the peak in concrete terms, the +6h bearish coefficient means a fully confident bearish article implies roughly 41% more volume than a neutral hour, since $\exp(0.342) - 1 \approx 0.41$. The shape is not a smooth exponential decay. There are secondary maxima at +1h and +4h, which points to discrete information-incorporation events at particular horizons rather than a continuous fade.

RQ2 (asymmetry). Bearish articles get a stronger response than bullish ones at the lags that matter (0, 1, 4, and the +6h peak, where bearish beats bullish by about 18%).

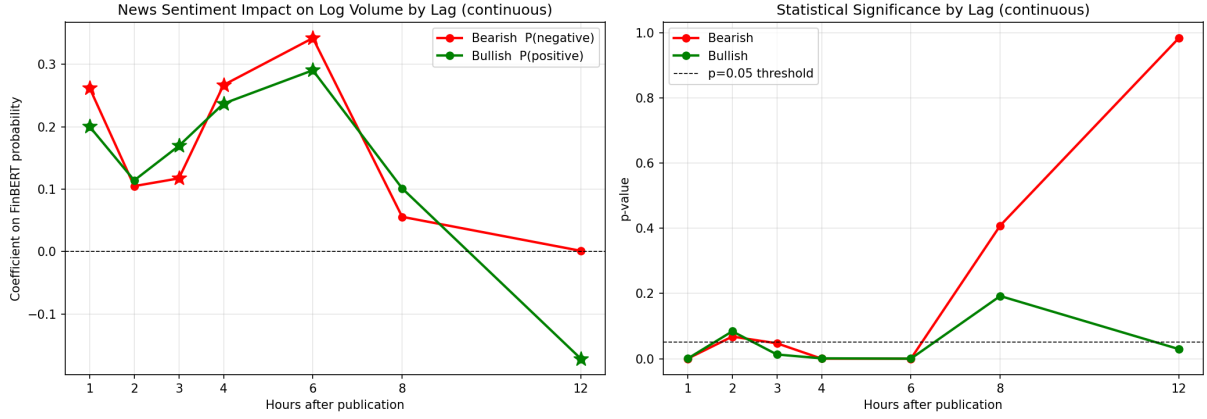


Figure 4.1: Lag OLS coefficients and p-values across horizons. Left panel: the bearish $P(\text{negative})$ and bullish $P(\text{positive})$ coefficients at lags 1, 2, 3, 4, 6, 8, and 12 hours. Stars mark lags significant at the conventional $p < 0.05$ threshold. Right panel: the p-values for the same coefficients, with the $p = 0.05$ reference line. The peak at lag +6h is the canonical finding for RQ1; the ordering of bearish above bullish at the dominant lags is the canonical finding for RQ2.

Lag 3 is the exception, with both coefficients small and bullish marginally ahead, but it is a weak lag and it does not overturn the pattern at the horizons carrying the effect. Since the two probabilities are estimated as independent regressors (Section 3.5.1), the bearish-over-bullish ordering is something the data produced, not something the model was set up to find.

On the small R^2 and robustness. R^2 never gets above 0.003. That is what we expected. Hourly volume has many drivers and sentiment is one of them, so R^2 here measures sentiment’s share of *total* volume variance, not of *news-driven* volume. The finding is the structured coefficient pattern, not the explained variance. Two robustness notes. The regression is cross-sectional, one article per row with no strong serial correlation, so we do not apply HAC corrections (Section 3.5.1). And the ceiling alignment guarantees every article precedes its matched hour, which if anything understates the immediate reaction, so the bias runs conservative for the peak-lag finding. Phase 2 revisits both RQ1 and RQ2 independently through the TFT v2 attention pattern (Section 4.3.7).

4.2.4 Phase 1 summary

Phase 1 produces the canonical statistical evidence for both research questions, and it exposes the limitations Phase 2 goes on to address.

Evidence. RQ1 gets its answer from the lag OLS peak at +6h, and RQ2 from the **bearish** > **bullish** ordering at the dominant lags (Section 4.2.3). Both are the primary Phase 1 evidence, and the TFT v2 attention pattern revisits both independently in Phase 2. The headline-bias experiment (Section 4.2.2) adds a separate finding: title-only and title-plus-body FinBERT disagree on 41.6% of articles, with titles leaning more bullish than the

bodies they head, which is what motivates extracting features from full bodies in Phase 2.

VAR exploration (abandoned). We fitted an exploratory VAR and then abandoned it. More than half the hours carry no contemporaneous news, so the sentiment series is mostly zeros and the impulse responses came back insignificant. The failure is still informative: it tells us the data are sparse and event-driven, and that property is exactly what motivates the deep-learning model in Phase 2.

Limitations that drive Phase 2. Five limitations of the Phase 1 pipeline each map onto a Phase 2 change (Table 4.5).

Table 4.5: Phase 1 limitations and the Phase 2 responses.

Phase 1 limitation	Phase 2 response
FinBERT reduces each article to one sentiment axis (no magnitude, event type, certainty, horizon, entities)	Structured LLM extraction with a richer schema (Section 4.3.2)
The regex <code>body_valid</code> filter is fast but not semantically discriminative (false positives and negatives)	LLM-judged <code>usable</code> flag (Section 4.3.3)
The sentiment signal is sparse ($\sim 50\%$ of hours have no news), defeating models that need joint observation	A TFT built for sparse, event-driven inputs (Section 4.3.5)
The corpus is pre-war only (ends February 2026)	Corpus extended through May 2026 (Section 4.3.1)
The OLS has no macro controls	DXY and VIX added as exogenous covariates (Section 4.3.5)

Every Phase 2 step answers a concrete limitation the previous step exposed, so the progression can be traced rather than just claimed.

4.3 Phase 2: refined pipeline and deep-learning modeling

Phase 2 takes on the Phase 1 limitations (Section 4.2.4) with a coordinated redesign: FinBERT out and structured LLM extraction in, DXY and VIX added as macro covariates, the regex filter replaced by an LLM-judged `usable` flag, a Temporal Fusion Transformer as the primary model, and the corpus extended through May 2026. It happened in a particular order, and the write-up follows that order. An initial LLM schema (v1) supported the first TFT training (TFT v1). An inter-model calibration then exposed a weakness in the composite sentiment score, which motivated a revised schema (v2) with three orthogonal economic channels and the `usable` flag. The full extraction then ran on the expanded corpus and supported the final TFT v2. So the subsections go: data

setup (Section 4.3.1), LLM extraction (Section 4.3.2), filter migration (Section 4.3.3), first calibration (Section 4.3.4), TFT v1 (Section 4.3.5), the schema response (Section 4.3.6), TFT v2 (Section 4.3.7), and the comparison of the two phases (Section 4.3.8).

4.3.1 Phase 2 data and feature setup

The Phase 2 corpus extends Phase 1 two ways: the GDELT scrape runs through May 2026 to cover the post-war period, and five new queries go after geopolitical and demand-side themes (Section 3.1). The full *filtered* corpus is **22,795 article records** from January 2024 to May 2026. Of those, **19,619 have substantive body content** (non-null, not a scraping error) and go to LLM extraction. The remaining 3,176 split two ways: 2,749 returned no body at all, and 427 returned text that was an error page rather than an article, the usual causes being paywalls, Cloudflare blocks, fetch failures, and pages needing JavaScript to render.

Macro covariates. We added hourly DXY (DX-Y.NYB) and VIX ($\sqrt{\text{VIX}}$) series as exogenous controls (Section 3.6.2). The reason is empirical: the Phase 1 OLS explained under 0.3% of volume variance (Section 4.2.3), so letting the model soak up broad market-driven variation should leave a cleaner residual for news to explain. We also moved the pipeline off CSV intermediates and onto the SQLite database, which makes the alignment a deterministic join.

Aligned dataset. Re-running the temporal alignment (Section 3.2) over the expanded corpus gives **22,795 article-aligned rows** (15,290 contemporaneous, 7,505 forward-assigned) on the 11,232-hour market grid from 13 May 2024 to 13 May 2026. Table 4.6 puts the two phases side by side.

Everything else in this chapter runs on the Phase 2 dataset.

4.3.2 LLM feature extraction

Swapping FinBERT for LLM extraction is the central Phase 2 change. It recovers the dimensions a single sentiment axis throws away: magnitude, event type, certainty, time horizon, entities, and the economic channels. Sections 3.3.2 through 3.3.5 specify the methodology. This subsection is about what happened when we ran it.

Extraction setup. Claude Haiku 4.5 scored each of the 19,619 articles with a substantive body, through the tool-use API, which forces the `extract_article_features` schema (Section 3.3.3) and lets us parse the tool call deterministically. The user message is the title and body, truncated to 1,500 characters, and Appendix A has the full prompt and tool schema. The schema short-circuits unusable articles down to just the `usable` flag, which cuts output tokens on the roughly 40% unusable share, and all 19,619 articles

Table 4.6: Phase 1 versus Phase 2 dataset.

Stage	Phase 1	Phase 2
GDELT queries	8	13
Article window	March 2024 – February 2026	January 2024 – May 2026
Raw articles (GDELT scrape)	51,948	76,345 (cumulative; Phase 2 re-scrape added 24,397)
After deduplication + English	16,326	22,795
Body retrieved	~13,000	20,046
Passing the regex filter (<code>body_valid=1</code>)	7,755	13,550
Sent to LLM extraction	n/a	19,619 (non-null, non-error body)
Title-only fallback	5,935	0 (no title-only fallback in Phase 2)
Modeling-ready corpus	13,690 (Phase 1 OLS/VAR)	10,514 (<code>usable_strict</code>) / 11,675 (<code>usable</code>)
Market hours window	March 2024 – February 2026	May 2024 – May 2026
Hourly market observations	11,219	11,232
Macro covariates	none	DXY, VIX
Sentiment representation	FinBERT 3-class probabilities	LLM continuous sentiment + 3 channels + categorical fields

returned a valid response with zero unrecoverable errors. The deployed schema is v2 (Section 3.3.4): the composite `sentiment_score`, three orthogonal channels (`supply_impact`, `demand_impact`, `risk_premium`), `magnitude`, `certainty`, `event_type` as an array of one to three categories, `time_horizon`, `entities`, and the required `usable` flag. The batch filtered on non-null, non-error bodies rather than the Phase 1 regex, which makes the LLM’s own `usable` flag the downstream filter. Section 4.3.3 compares it against the regex.

Extraction outcome. The LLM flagged **11,675 articles as usable (59.5%)** and 7,944 as unusable (40.5%, mostly paywalls, Cloudflare blocks, and keyword-matched off-topic content). That lines up with both the pre-batch estimate and the calibration sample (Section 4.3.4). Table 4.7 gives the four continuous scores across the 11,675 usable articles.

Table 4.7: Continuous feature means over the 11,675 usable articles.

Feature	Mean	Std
<code>sentiment_score</code>	+0.025	0.501
<code>supply_impact</code>	−0.071	0.463
<code>demand_impact</code>	−0.075	0.340
<code>risk_premium</code>	+0.156	0.426

The near-zero `sentiment_score` mean says the corpus is roughly balanced between bearish and bullish. The negative `supply_impact` and `demand_impact` means say it tilts toward supply-tightening and demand-weakening news. And the positive `risk_premium` mean, the largest of the four in magnitude, reflects how elevated geopolitics were over the coverage period. One number worth flagging: 9.9% of usable articles have all three channels at exactly zero, meaning the LLM called them usable but materially neutral. Section 4.3.4 comes back to this, and Section 4.3.3 uses it to define the stricter `usable_strict` filter.

Categorical and entity distributions. First-rank event types are mostly geopolitical (37.6%) and supply (28.6%), then macro (14.5%) and demand (12.7%). After normalisation onto the canonical set, the most frequent entities are the United States (4,487), Iran (3,202), Trump (1,935), Russia (1,744), China (1,588), OPEC+ (1,382), Israel (1,100), India (956), OPEC (845), and the Strait of Hormuz (832). All of these make sense for the coverage period. The normalisation step is doing real work: in the raw extraction “United States” and “US” appear as separate strings (3,092 and 727 mentions), and the alias mapping merges them together with the remaining variants. That the Strait of Hormuz survives as its own entity, at a count close to OPEC’s, is evidence the mapping separates chokepoints from actors rather than folding them into Iran. These distributions are the input for the TFT models below.

4.3.3 LLM usable flag and filter comparison

The Phase 1 regex `body_valid` filter (Section 3.4.1) and the Phase 2 LLM `usable` flag (Section 3.4.2) are two answers to the same question: which scraped articles carry substantive content? Both labels exist for the 19,619 processed articles, so we can just compare them.

Result. Table 4.8 is the 2×2 contingency table.

Table 4.8: Regex versus LLM filter agreement (19,619 articles).

	LLM usable=0	LLM usable=1	Total
regex <code>body_valid</code> =0	4,578	1,491	6,069
regex <code>body_valid</code> =1	3,366	10,184	13,550
Total	7,944	11,675	19,619

The regex accepts 69.1% of articles and the LLM 59.5%, and they agree on 75.2% of them (14,762 of 19,619), which is a **Cohen’s κ of 0.47**, chance-corrected, or “moderate” on the Landis-Koch convention. The interesting part is the two off-diagonal cells. The regex rejects but the LLM accepts **1,491** articles, the regex’s false negatives, which are short but substantive pieces sitting under the 400-character threshold. The regex accepts but the LLM rejects **3,366**, the regex’s false positives, which are long keyword-matched articles

that are not about WTI. So the regex is too permissive on long non-substantive content, because it cannot judge topic semantically, and too restrictive on short substantive content, because its length cutoff is blunt. The LLM, which can read the body, gets both of these the other way round.

Manual audit. We sampled both off-diagonal cells and hand-labelled them, and the LLM’s judgment holds up in the large majority of cases. The regex’s false positives are mostly keyword collisions: palm oil, canola, a semiconductor firm with an energy-sounding name, generic equity wraps. Its false negatives are short but on-topic briefs, for instance a 320-character note on OPEC+ holding its production plan. The audit also turned up a residual LLM error mode. A few articles the LLM marks `usable` are not really WTI-relevant, the clearest being one about Canada rolling back EV tariffs. So the LLM filter beats the regex on both of the regex’s failure modes without being a ground-truth oracle itself, a limitation Chapter 5 returns to.

The `usable_strict` variant. A distinct group of articles is marked `usable` and yet comes back with all three channel scores at exactly zero: **1,161** of them, the 9.9% from Section 4.3.2. We define a stricter filter for these, `usable_strict=1`, requiring at least one non-zero channel, which leaves **10,514** articles. It is the canonical filter for TFT v2 training and the reported ablation (Appendix C), while `usable=1` is only used for this filter comparison.

Our first hypothesis was that these would turn out to be the LLM’s residual false positives, the EV-tariffs kind from the audit above, finally showing themselves through a diagnostic pattern. A manual audit did not confirm it, and the distinction matters for how the filter should be understood. Inspecting the zero-channel articles turned up no off-topic contamination: no EV stories, no scraping artifacts, nothing resembling the keyword collisions that characterise the regex’s false positives. They are a mix of two things instead. Some are genuine residual punting, where the LLM tagged the article with a channel-relevant event type (supply, demand, or geopolitical) and then declined to commit to a magnitude on the matching channel, which is the same conservatism the calibration flagged in Section 4.3.6. The rest are on-topic articles whose channels are correctly zero for what they are: pure price reports, technical analysis, equity-market reactions to oil moves, and macro framing with no specific supply or demand implication. Nothing is wrong with that second group.

So `usable_strict` is not a correctness filter and we do not claim it as one. It is a signal filter. The channels are what Phase 2 contributes, and an article carrying no channel signal contributes nothing to them regardless of whether it is a good article, so training on it adds noise to the features the model is meant to learn from. The cost of the choice is that roughly a thousand substantive articles are dropped for a property that is a correct description of

them, which is a real trade-off rather than a free improvement, and Appendix C tests the `usable` against `usable_strict` configurations directly.

Filter choice and cross-phase note. The two filters are not nested. The LLM accepts 1,491 articles the regex rejects and rejects 3,366 it accepts, so its acceptance set overlaps the regex’s without being a subset of it. The LLM filter is the canonical Phase 2 downstream filter. Between this migration and the loss of Phase 1’s title-only fallback, the Phase 1 modeling set (13,690) and the Phase 2 set (11,675 usable, 10,514 strict) are not directly comparable, which Section 4.3.8 develops.

4.3.4 Inter-model calibration of LLM features

Since the LLM-extracted features are what all of Phase 2 sits on, we validate them before using them. Following Section 3.7, we run an **inter-model calibration** [Gilardi et al., 2023, Zheng et al., 2023]: the same 30 stratified articles get scored by Claude Haiku 4.5 and OpenAI’s GPT-5 with an identical prompt and schema, and we treat cross-family disagreement as a signal of either genuine ambiguity or systematic bias. This subsection reports the first calibration, on Schema v1, and what it told us.

Result on Schema v1. Schema v1 had a single composite `sentiment_score` as its directional feature. The two models agreed on the `usable` flag for 26 of 30 articles (87%), but agreement on the composite sentiment was bad: a Pearson correlation of only **0.39**, with **opposite-signed** scores on **4 of 13 usable articles (31%)**. Sign disagreement is the worst kind to have, because it means the two models read the same article as pushing the WTI price in opposite directions.

Diagnostic (the Phase 2 pivot). The sign disagreements are not evenly spread. The two highest-magnitude articles in the sample, both scored 0.75 by Haiku, disagree on sign, and both are geopolitical supply stories: an Iran escalation warning and the UAE announcing its exit from OPEC. The other two disagreements sit at the opposite end, on low-magnitude single-company stories (0.15 and 0.25, on Occidental and Baytex results), where both models produced small opposite-signed values. It is the high-magnitude pair that matters, because those are the articles that move the market. The cause is that the composite `sentiment_score` is being asked to hold two separate judgments at once: the **valence of the event** (is this good news or bad news?) and its **directional price impact** (does it push crude up or down?). For most articles those agree, so nothing breaks. For a supply-threatening escalation they come apart: the event is negative in valence but bullish for price. A model weighting valence scores it negative, a model weighting price scores it positive, and one number cannot hold both. So the two models resolve the ambiguity differently exactly where it matters most. A 0.39 correlation on the primary feature is a structural problem, not bad luck.

Response. The fix is to decompose the judgment rather than fiddle with the prompt. Section 4.3.6 introduces the three orthogonal economic channels (`supply_impact`, `demand_impact`, `risk_premium`) and the re-calibration that validates them. But TFT v1 was trained on the v1 schema before any of that, so its results and limitations come first.

4.3.5 Temporal Fusion Transformer (TFT) v1

TFT v1 is the project’s first deep-learning model and the first to use the richer LLM features. It trained on the Phase 1 corpus, with Schema v1 features plus the DXY and VIX macro covariates. The Temporal Fusion Transformer [Lim et al., 2021] is an attention-based multi-horizon forecaster, and its Variable Selection Network (VSN) and interpretable attention are what make it a good fit here.

Training setup. The architecture (Section 3.6) uses a 48-hour encoder, a one-hour horizon, `hidden_size=32`, `attention_head_size=4`, `dropout=0.1`, and a quantile loss, for about 113,000 parameters. On 10,797 hourly rows with an 80/20 temporal split, early stopping fired at epoch 21 with a best validation loss of **0.204**. The inputs are the Schema v1 LLM features (`sentiment_score`, `magnitude`, integer-encoded `event_type/price_direction/time_horizon`, `certainty`, `n_articles`), the market context (`log_volume`, `price_range`, `log_return`, `amihud`, `dxy`, `vix`), and calendar covariates.

Feature importance. One feature takes most of the VSN weight (Table 4.9).

Table 4.9: TFT v1 feature importance.

Feature	Importance	Group
<code>sentiment_score</code>	0.53	News
<code>log_volume</code>	0.095	Market
<code>dxy</code>	0.075	Market
<code>log_return</code>	0.055	Market
<code>event_type_num</code>	0.032	News
<code>magnitude</code>	0.024	News
<code>vix</code>	0.005	Market

Three things stand out. First, **`sentiment_score` dominates at 0.53, over five times the next feature**, which is a strong result for the LLM approach: the continuous LLM sentiment carries far more signal than any market variable, and far more than a FinBERT label could. Second, **VIX is negligible at 0.005** while DXY ranks third, so for WTI the dollar index is apparently already absorbing most of the macro-risk signal. Third, the categorical features (`event_type`, `price_direction`, `time_horizon`) were integer-encoded, which imposes a numeric order that means nothing and probably understates how informative they are. TFT v2 fixes this.

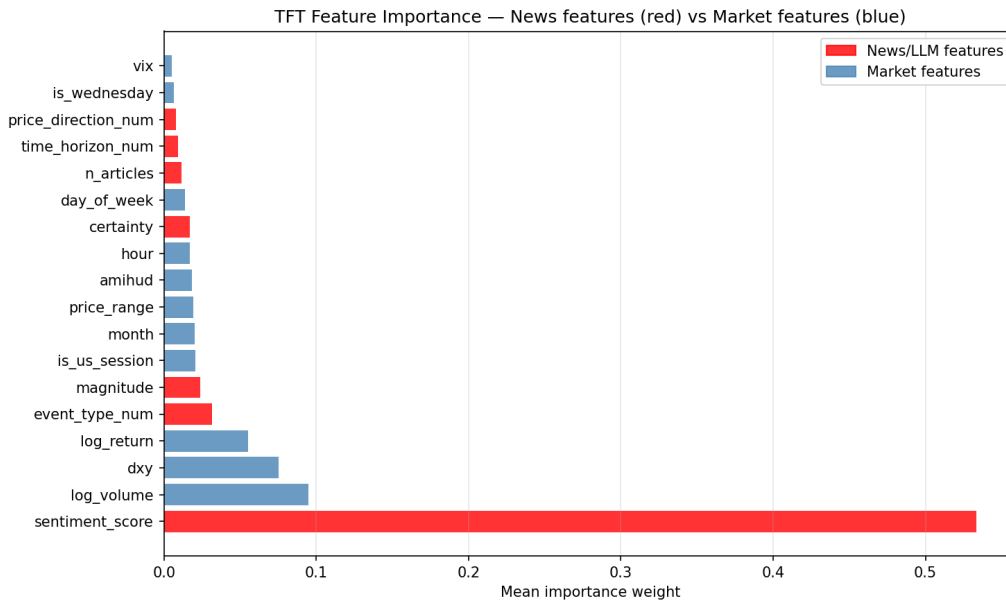


Figure 4.2: TFT v1 feature importance, news features (red) versus market features (blue). `sentiment_score` dominates at 0.53, more than five times the next feature; the market-context features form a second tier and the rest carry minor weight.

Attention and RQ1. Attention peaks at **lag $-4h$** , with a cluster from $-2h$ to $-5h$, which independently corroborates the Phase 1 lag OLS peak at $+6h$ (Section 4.2.3): two very different methods both put the dominant news-impact horizon several hours after publication. There is also a secondary cluster at $-27h$ to $-28h$, a “daily memory” effect at the same hour on the previous trading day, which the lag OLS could never have seen because it stops at $12h$.

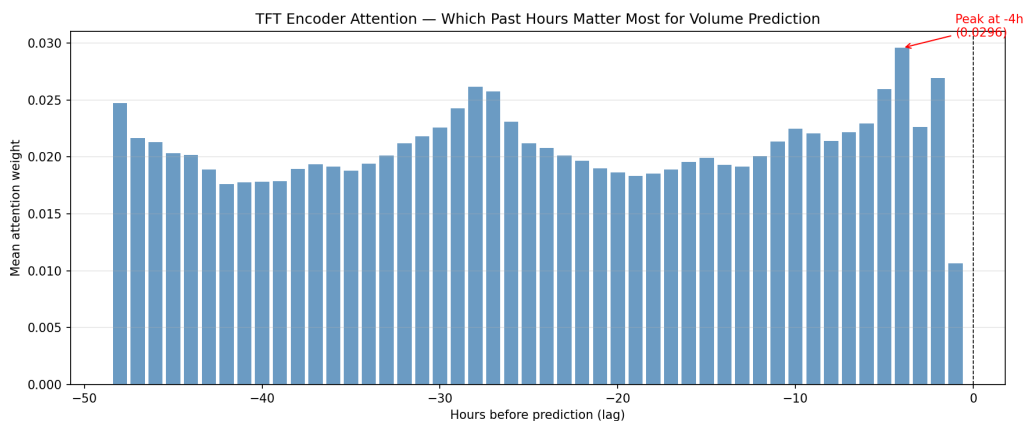


Figure 4.3: TFT v1 encoder attention by lag. Mean attention over the 48-hour encoder window, peaking at $-4h$. Two clusters are visible: a short-term one between $-2h$ and $-5h$, and a secondary one around $-27h$ to $-28h$.

Directional asymmetry (RQ2). Splitting validation samples by sentiment direction gives the mean predicted `log_volume` in Table 4.10.

Bearish predicted volume comes out above bullish, which points the same way as Sec-

Table 4.10: TFT v1 mean predicted `log_volume` by sentiment direction.

Sentiment direction	Mean predicted <code>log_volume</code>
Bearish	8.775
Bullish	8.700
Neutral	8.875

tion 4.2.3, but a two-sample t-test on the difference is not significant ($p = 0.56$): the validation split holds only 156 bearish and 144 bullish hours, which is not enough to power the test. So the lag OLS stays the canonical RQ2 evidence and the TFT only corroborates the direction. Neutral hours have the highest predicted volume of the three, which is an “uncertainty drives trading” story we flag as speculative and leave for the discussion.

Limitations of TFT v1 that drive TFT v2. TFT v1 validates the approach: LLM sentiment dominates, attention corroborates the lag OLS, and the asymmetry shows up in direction. But five limitations each motivate a Phase 2 change:

1. **Underpowered asymmetry test** ($p = 0.56$), motivating a larger corpus.
2. **Integer-encoded categoricals** impose a meaningless numeric order, motivating proper categorical encoding.
3. **No entity awareness** (entities extracted but unused), motivating 71 entity flags.
4. **Pre-war training data only**, motivating a corpus extended through May 2026.
5. **The dominant feature is the unreliable composite sentiment:** the 0.39-correlation feature from Section 4.3.4 carries 53% of the model’s weight, motivating the channel decomposition.

The fifth one is the pivot. The model leans hardest on exactly the feature the calibration flagged as least trustworthy, so v2 replaces it with the orthogonal channels (Section 4.3.6) and retrain (Section 4.3.7).

4.3.6 Channel decomposition response

The calibration (Section 4.3.4) showed the composite `sentiment_score` was unreliable ($r = 0.39$, 31% sign disagreement) on exactly the high-magnitude geopolitical events that matter most, and TFT v1 was putting 53% of its weight on it. This subsection is the schema revision that fixes that, and the re-calibration that checks the fix.

Design principle. An oil-market article carries up to three economically distinct judgments: what it implies for **supply**, for **demand**, and for **geopolitical or operational risk**. That mirrors the standard structural separation of oil-price drivers into supply, aggregate-demand, and precautionary (risk) components [Kilian, 2009]. A single composite

forces all three into one number, and when they point different ways it breaks. A supply shock is bearish for the event but bullish for price. A de-escalation is positive in valence but bearish for supply through sanctions relief. Schema v2 therefore splits the judgment up so each score carries one factual claim.

Schema v2 changes. The main addition is the three orthogonal channels, each on $[-1, +1]$: `supply_impact` (positive = more oil to market), `demand_impact` (positive = stronger consumption), and `risk_premium` (positive = elevated risk). Schema v2 also promotes `usable` to a required field, makes `event_type` a salience-ordered array, drops the redundant `price_direction`, and keeps the composite `sentiment_score`. Appendix B has the full field-by-field v1-vs-v2 comparison.

Re-calibration. Re-scoring the same 30-article sample with Schema v2 improves every metric (Table 4.11).

Table 4.11: Inter-model re-calibration metrics, Schema v1 versus v2.

Metric	v1 schema	v2 schema
<code>usable</code> agreement	26/30 (87%)	27/30 (90%)
<code>sentiment_score</code> correlation	0.39	0.88
<code>sentiment_score</code> sign disagreements	4/13 usable (31%)	1/14 usable (7%)
<code>supply_impact</code> correlation	—	0.94
<code>supply_impact</code> MAD	—	0.11
<code>supply_impact</code> sign disagreements	—	0
<code>demand_impact</code> correlation	—	0.96
<code>demand_impact</code> MAD	—	0.05
<code>demand_impact</code> sign disagreements	—	0
<code>risk_premium</code> correlation	—	0.82
<code>risk_premium</code> MAD	—	0.13
<code>risk_premium</code> sign disagreements	—	1

The headline numbers: `sentiment_score` correlation goes from **0.39 to 0.88** and its sign disagreements from **31% to 7%**, and every new channel correlates at 0.82 or better (`supply_impact` 0.94, `demand_impact` 0.96, `risk_premium` 0.82) with near-zero sign disagreements. The surprising part is that the composite improved even though we never touched it. Asking the model to decompose the article first and only then produce a composite seems to discipline everything it outputs. Within the calibration sample the channels are close to orthogonal, with pairwise $|r| \leq 0.40$ across Haiku’s scores on the 14 both-usable articles, so they are capturing distinct economic dimensions rather than the same judgment under three names. On the full corpus the picture is slightly less clean. `demand_impact` is essentially uncorrelated with the other two ($|r| < 0.05$), but `supply_impact` and `risk_premium` correlate at -0.55 . That is economically expected rather than a defect, since a supply threat is itself a risk event, and it is the same coupling

the geopolitical findings of Section 4.3.7.4 rest on.

Residual weakness. On 3 of the 14 usable calibration articles, Haiku zeroed all three channels where GPT gave small non-zero values, on articles of modest relevance. The matching `sentiment_score` and `magnitude` were subdued too, so these do not dominate anything downstream, but this conservatism is the same pattern the `usable_strict` filter is built on (Section 4.3.3). With the channels validated, the full Schema v2 batch of 19,619 articles produces the feature set for TFT v2 (Section 4.3.7).

4.3.7 Temporal Fusion Transformer v2

TFT v1 showed the deep-learning approach works. TFT v2 brings together everything Phase 2 invested in (the channel decomposition of Section 4.3.6, the LLM filter of Section 4.3.3, the expanded corpus, and entity normalization) and is the model we evaluate the research questions against. A controlled ablation guided the design (Appendix C). The configuration reported here is not the most accurate one at every horizon: the simpler single-target variants predict the one-hour horizon better (test MAE 0.424 against 0.585). It is the configuration that answers the research questions, because only the multi-target, multi-horizon, entity-aware model yields the per-horizon error curve and the direction-conditioned attention the analysis depends on. Appendix C sets out that trade explicitly.

4.3.7.1 Success criteria

The three criteria reflect what Phase 2 is for. The Phase 1 OLS gives the primary evidence for RQ1 (the +6h peak) and RQ2 (bearish > bullish), and TFT v2’s job is to corroborate that through an independent method while showing the channels and entities as visible drivers.

1. **Channels and entities are visible drivers.** The VSN importance ranking should place channel features and entity flags in the top ranks, with a non-degenerate, interpretable attention pattern. *The channel decomposition and the entity normalisation are the two feature investments of Phase 2. If they carry genuine market information, the Variable Selection Network is where that has to become visible; a model that ignored them would indicate the investment bought nothing.*
2. **Prediction substantially beats persistence** in the 1-to-12h window, confirming the Phase 2 features carry predictive signal. *Persistence is a deliberately low bar, and clearing it is the minimum evidence that the features carry usable signal rather than merely interpretable weights. A model that ranks features sensibly but cannot beat a naive carry-forward has not learned anything about liquidity.*

3. **The multi-horizon error curve is consistent with Phase 1’s lag structure**, with the largest persistence reduction in the +6h range identified by the lag OLS. *Agreement with Phase 1 is the reason for running a second method at all. If a nonlinear forecaster trained on different features and a wider corpus locates the response in the same window as a sequence of single-lag linear regressions, that window is unlikely to be an artefact of either method.*

Section 4.3.7.7 has the evaluation.

4.3.7.2 Model design

TFT v2 uses a hidden size of 32, four attention heads, dropout 0.15, and a hidden-continuous size of 16, for 298,329 parameters. It trains on the `usable_strict=1` filter (Section 4.3.3), so on 10,514 articles, with the 1,161 channel-neutral ones excluded. The input features are:

- **Channels (Phase 2 contribution):** `supply_impact`, `demand_impact`, `risk_premium`, each on $[-1, +1]$.
- **Composite news:** `sentiment_score`, `magnitude`, `certainty`, `n_articles`.
- **Categoricals:** `event_type_primary` (8 categories including the synthetic `no_news`) and `time_horizon` (4), as proper embeddings (dimension 5 and 3), taking the dominant article’s value (highest `magnitude`, tie-broken on `article_id`).
- **Entity flags (Phase 2 contribution):** 71 binary multi-hot columns (Section 4.3.2).
- **Market context:** `log_volume`, `price_range`, `log_return`, `amihud`, `dxy`, `vix`.
- **Calendar:** `hour`, `day_of_week`, `month`, `is_us_session`, `is_wednesday`.

Continuous features are hour-averaged, entity flags take the maximum, and categorical features take the highest-magnitude article. Hours with no news get the `no_news` category and zeros. A few boundary nulls (DXY/VIX holiday gaps, first-hour `log_return`/`amihud`) are forward-filled or set to zero, which affects under 0.1% of samples.

The model predicts three targets jointly (`log_volume`, `amihud`, `price_range`) at four horizons (**1, 3, 6, 12h**), and the +6h horizon is there to match the lag OLS peak (Section 4.2.3). The temporal split is **60/20/20** on the 11,232 hourly observations, with 48h buffers between partitions: train 2024-05-13 to 2025-06-11, validation 2025-06-13 to 2025-12-27 (2,216 samples), test 2025-12-29 to 2026-05-13 (2,159). The war onset (28 February 2026) lands inside the test window, splitting it into 994 pre-war and 1,165 war hours, so the test set measures how a pre-war-trained model generalizes to an unseen war regime. We report metrics on the full test set, the pre-war slice, and the war slice throughout.

Training uses Adam (learning rate 1e-3, halved after three epochs without validation improvement). The objective is a quantile loss applied to each of the three targets independently and then summed, so the model learns a spread of quantiles (a prediction interval) per target rather than a single point estimate. Each target is normalized separately, with normalization statistics pooled within the asset group, which here is just WTI, so the three targets sit on comparable scales before their losses combine. Gradients are clipped at 0.1, training stops early once validation loss flattens (patience 10, minimum delta 1e-4), and the run is seeded (42) with deterministic algorithms on a Colab T4 GPU. Best validation loss was **0.427** at epoch 10.

4.3.7.3 Predictive performance

We evaluate TFT v2 on the held-out test set for all three targets at all four horizons, against a persistence baseline (the current value carried forward, the standard financial-forecasting reference), reporting the median quantile (q50) on four slices: validation, full test, the pre-war test portion (994 hours before the 28 February 2026 onset), and the war portion (1,165 hours from onset to the end of the data).

Table 4.12: `log_volume`: TFT v2 versus persistence MAE.

Horizon	Persistence MAE	TFT v2 MAE	Reduction	Pre-war MAE	War MAE
1h	1.076	0.585	46%	0.536	0.628
3h	1.452	0.577	60%	0.537	0.611
6h	1.820	0.602	67%	0.551	0.646
12h	2.174	0.631	71%	0.568	0.685

The error reduction over persistence climbs steadily with horizon, from 46% at 1h to 71% at 12h, which happens because persistence gets worse at longer horizons as volume autocorrelation decays. Since the growth is driven by the baseline rather than by the model, this curve should not be read as locating the news response at +12h; Section 4.3.7.5 takes that point up in full. Error runs 14 to 21% higher on the war slice than pre-war at every horizon, which is the price of extrapolating from pre-war-only training, but it is small enough that the model still has real predictive value on a regime it never saw.

Table 4.13: `amihud`: TFT v2 versus persistence MAE.

Horizon	Persistence MAE	TFT v2 MAE	Reduction	Pre-war MAE	War MAE
1h	0.0004	0.00023	43%	0.00010	0.00035
3h	0.0004	0.00023	43%	0.00010	0.00034
6h	0.0004	0.00022	45%	0.00009	0.00034
12h	0.0004	0.00022	45%	0.00009	0.00033

Amihud gives a steady 43 to 45% reduction across horizons with little variation, though

part of that flatness is an artifact of amihud’s compressed $1e-4$ scale. War-slice persistence MAE is three to four times the pre-war value, which reflects illiquidity spikes in the war regime, but the model’s relative reduction over persistence holds up on both slices.

Table 4.14: `price_range`: TFT v2 versus persistence MAE.

Horizon	Persistence MAE	TFT v2 MAE	Reduction	Pre-war MAE	War MAE
1h	0.495	0.718	−45%	0.154	1.200
3h	0.578	0.719	−24%	0.156	1.200
6h	0.630	0.721	−14%	0.161	1.198
12h	0.701	0.724	−3%	0.165	1.201

This is where the model loses. It underperforms persistence on the full test set, by -3 to -45% , and the failure is specific to a regime rather than general. On the pre-war slice the model does well: its error runs 0.154 to 0.165 and it beats persistence there rather than merely matching it (Appendix F). On the war slice it degrades to about 1.200, which is worse than war-slice persistence at 0.764 to 1.055. Having only ever seen the moderate-volatility pre-war regime, the model reverts toward the historical mean when asked to predict the war regime, while persistence at least tracks the elevated current state. This is a clean regime-extrapolation failure, and it makes sense that `price_range` is where it shows up: it measures intraday volatility directly, so it is far more regime-sensitive than the aggregate targets. Chapter 5 takes up the implications.

Across targets. The model gets substantial gains on `log_volume` (46 to 71%) and `amihud` (43 to 45%) and fails on `price_range` in the war regime, which fits the train/test regime mismatch: the aggregate liquidity measures transfer across regimes and `price_range` does not. So the rest of Section 4.3.7 concentrates on `log_volume`, which is both the target most relevant to RQ1 and RQ2 and the one the model handles best. The `price_range` failure comes back in Section 4.3.7.5 and in Chapter 5.

4.3.7.4 Feature contributions

The Variable Selection Network (VSN) gives each input feature a learned weight for how much the model relies on it, and aggregating those weights over validation samples ranks the features. Table 4.15 has the top ten for `log_volume` prediction.

Three patterns come out of this. First, VIX ranks first at 18.8%, which fits its role as a market-wide volatility proxy: `log_volume` tracks broad equity-market activity through stress periods, and risk expectations shape how much volume responds to oil news. Second, the channel decomposition puts two features in the top five, `supply_impact` at 12.1% and `demand_impact` at 5.5%, which together are 17.6% of importance, roughly matching VIX on its own. The composite `sentiment_score` ranks 21st of the 90 features at 0.9%, below

Table 4.15: TFT v2 top ten features by VSN importance (`log_volume`).

Rank	Feature	Importance	Type
1	<code>vix</code>	0.188	macro covariate
2	<code>supply_impact</code>	0.121	channel (news)
3	<code>ent_oman</code>	0.113	entity flag (news)
4	<code>demand_impact</code>	0.055	channel (news)
5	<code>is_wednesday</code>	0.022	calendar
6	<code>ent_japan</code>	0.017	entity flag (news)
7	<code>ent_eu</code>	0.016	entity flag (news)
8	<code>ent_iran</code>	0.015	entity flag (news)
9	<code>ent_china</code>	0.014	entity flag (news)
10	<code>ent_algeria</code>	0.014	entity flag (news)

`risk_premium` (19th) and `log_volume` (20th), which suggests that once the model has the decomposed channels the aggregate sentiment has little left to add.

Third, six of the top ten are entity flags, and the entity block as a whole carries **52%** of total VSN importance across its 71 flags, more than the channels and the macro covariates combined. That aggregate is the substantive finding: which actors a story names matters more to the model than the direction the story implies. The ordering *within* that block should not be over-read, for two reasons. Corpus frequency does not predict a flag’s weight, with a Spearman ρ of 0.13 ($p = 0.28$) across the 70 flags whose article counts can be recovered from the entity table, so the leading flags are not simply the most-discussed actors: `ent_oman` appears in 135 articles and ranks 3rd, while `ent_us` appears in 4,487 and ranks 29th. And a geographic reading, on which Oman, Iran and Qatar all sit around the Strait of Hormuz, does not survive checking, because the `ent_strait_of_hormuz` flag itself ranks 87th of 90 and `ent_saudi_arabia` 81st. `ent_oman` is also an outlier in magnitude, at nearly seven times the next entity flag and 22% of the entire entity block on a flag that fires in 1.3% of articles, which makes it the least stable entry in the table and the one the retraining check below displaces first.

The two choices behind this result, proper categorical encoding and the entity flags, were checked in a controlled three-variant ablation. Its key finding is that the channels’ predictive role only appears under proper categorical encoding, where `demand_impact` becomes the top feature, and not under the integer encoding of the simplest variant, where the composite sentiment score dominates instead. The full three-variant comparison is in Appendix C, to keep the main text on the canonical model.

The VSN tells us a feature’s weight, not the direction or size of its effect, so this reading is qualitative: VIX informs volatility expectations, the channels capture fundamental supply and demand news, and the entity flags carry source-specific risk signals. Sections 4.3.7.5 and 4.3.7.6 look at how the model turns these features into predictions, through its

attention pattern and its per-horizon error structure.

Stability of the ranking. The exact ordering in Table 4.15 does not survive retraining, and it is noting that. An independent run, retrained after adding two candidate covariates and a minor data-cleaning fix, reshuffled the ranking substantially: `risk_premium` and `magnitude` moved into the top two while `supply_impact` and `ent_oman` dropped out of it. What is stable across both runs is the character of the leading features rather than their order. Risk and volatility proxies (`VIX`, `risk_premium`) and news-salience measures lead in both, while the composite `sentiment_score` and the article count sit near the bottom in both. So these importance results should be read at the level of feature type, meaning risk and salience over price direction (Section 4.3.7.6 and Chapter 5), and not as a precise per-feature ranking.

4.3.7.5 Lag structure analysis: evidence for RQ1

RQ1 asks at what lag news events most strongly affect liquidity. Phase 1’s lag OLS (Section 4.2.3) put the bearish-news peak on `log_volume` at +6h ($\beta_{P(\text{neg})} = 0.342$), with the effect decayed to insignificance by +8h and secondary maxima at +1h and +4h. We test TFT v2 against that in two directions: the per-horizon error curve and the attention pattern.

Per-horizon error curve. If news impact really concentrates at a particular horizon, then a multi-horizon model should be most accurate there once we normalize for how hard each horizon is to begin with, which is what MAE reduction over persistence attempts. On `log_volume` (Table 4.12) that curve grows steadily from 46% at +1h to 71% at +12h, with +6h close behind at 67%.

This diagnostic is weaker than it first looks, and the reason is worth stating rather than glossing. The reduction grows mainly because the *baseline* deteriorates, not because the model improves: persistence MAE rises from 1.076 to 2.174 across those horizons while TFT v2’s own error moves only from 0.585 to 0.631. Most of the widening gap therefore reflects the decay of volume autocorrelation, which is a property of the target series, rather than any concentration of news impact at +12h. Read strictly, the curve cannot locate the response on its own, and it does not corroborate the +6h peak by itself: Phase 1’s bearish coefficient at +12h is 0.001 at $p = 0.98$, so there is nothing there to agree with. What the curve does establish is the weaker but still useful claim that the Phase 2 features carry predictive signal across the whole 1-to-12-hour window, which is the second success criterion of Section 4.3.7.1. The lag-locating evidence has to come from the attention pattern below.

Attention pattern. The encoder attention distribution, aggregated over validation samples, gives a second and complementary view of the lag structure.

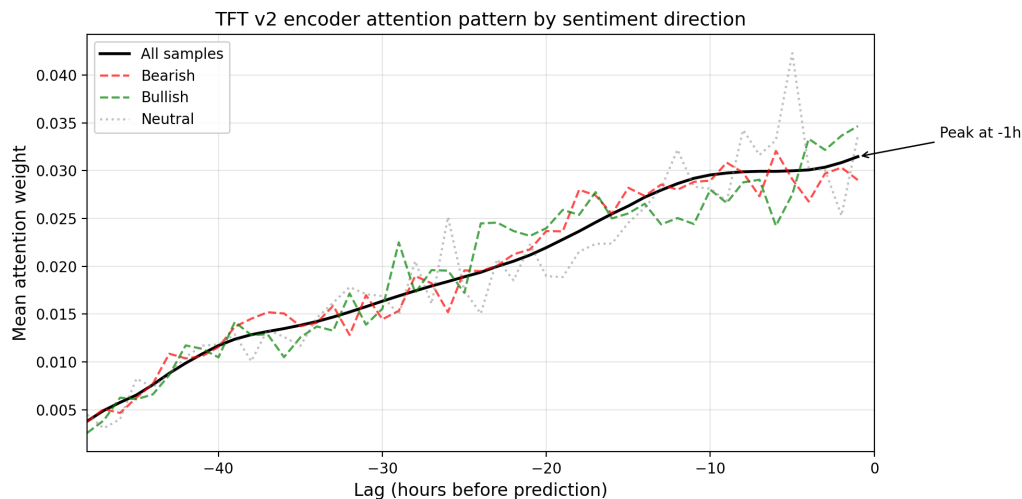


Figure 4.4: TFT v2 mean encoder attention across the 48-hour window, disaggregated by sentiment direction. Overall attention peaks at $-1h$; bearish-sentiment hours peak at $-6h$ while bullish-sentiment hours peak at $-1h$.

Figure 4.4 shows mean encoder attention across the 48-hour window. Overall attention peaks at lag $-1h$ (weight 0.0315) and falls off smoothly through $-2h$ to $-5h$ (0.0308, 0.0304, 0.0301, 0.0300), a graded recency effect with nothing piling up at the boundary. Split by sentiment direction, though, the pattern comes apart: bearish-sentiment hours peak at $-6h$ (0.0321) while bullish-sentiment hours peak at $-1h$ (0.0347). The bearish peak at $-6h$ lines up exactly with the $+6h$ lag OLS peak. That says the model has learned bearish news needs a longer integration window than bullish news, and it confirms independently the same structure Phase 1 found in the average bearish impact.

Synthesis on RQ1. The two diagnostics do not carry equal weight, and the honest summary distinguishes them. The attention pattern is the substantive corroboration: bearish-sentiment attention peaks at $-6h$, matching the lag OLS peak exactly, and it does so under a method that shares no assumptions with a single-lag linear regression. The per-horizon error curve is consistent with a response somewhere in the 1-to-12-hour window but, for the reason given above, cannot pin it to a lag. So the $+6h$ finding rests on two independent analyses, the Phase 1 regression and the Phase 2 attention, with the error curve as supporting rather than locating evidence.

4.3.7.6 Directional asymmetry analysis: evidence for RQ2

RQ2 asks whether bearish news produces a different liquidity response than comparable bullish news. Phase 1’s lag OLS (Section 4.2.3) found a robust bearish $>$ bullish asymmetry in $\log_volume$, strongest at $+6h$, and it remains the primary answer to RQ2. What we test here is whether TFT v2’s predictions add complementary evidence through an independent method.

Test design. At each horizon and slice we split samples by the sentiment score at the prediction hour into bearish (below -0.1), bullish (above $+0.1$), and neutral (within plus or minus 0.1), then test whether mean predicted `log_volume` differs between the bearish and bullish subsets using Welch’s t-test with unequal variance. Neutral hours are excluded because they capture low-news activity, dominate the sample counts, and dilute the contrast the test is built to find. That gives sixteen tests, four horizons times four slices (val, test_full, test_prewar, test_war), in Table 4.16.

Table 4.16: Directional asymmetry tests: mean predicted `log_volume`, bearish versus bullish.

Horizon	Slice	Bear. n	Bull. n	Bear. mean	Bull. mean	Diff.	p	Sig.
1h	val	320	271	8.345	8.281	+0.064	0.578	No
1h	test_full	384	826	8.928	8.857	+0.071	0.316	No
1h	test_prewar	180	340	8.853	8.783	+0.070	0.543	No
1h	test_war	204	486	9.001	8.901	+0.100	0.253	No
3h	val	320	271	7.938	8.131	-0.193	0.270	No
3h	test_full	384	826	8.732	8.596	+0.136	0.247	No
3h	test_prewar	180	340	8.706	8.365	+0.341	0.058	No
3h	test_war	204	486	8.756	8.729	+0.027	0.869	No
6h	val	320	271	8.186	8.369	-0.183	0.230	No
6h	test_full	384	826	8.633	8.529	+0.104	0.400	No
6h	test_prewar	180	340	8.470	8.380	+0.089	0.636	No
6h	test_war	204	486	8.788	8.615	+0.174	0.285	No
12h	val	320	271	8.277	8.371	-0.094	0.561	No
12h	test_full	384	826	8.220	8.464	-0.244	0.070	No
12h	test_prewar	180	340	8.274	8.333	-0.059	0.761	No
12h	test_war	204	486	8.169	8.540	-0.371	0.053	No

Results. None of the sixteen tests reaches significance at $p < 0.05$. The closest is +12h on test_war (bearish 8.169 vs bullish 8.540, difference -0.371 , $p = 0.053$). Two others get near without crossing: +3h test_prewar (bearish $>$ bullish, $+0.341$, $p = 0.058$) and +12h test_full (bearish $<$ bullish, -0.244 , $p = 0.070$).

Interpretation. This test and Phase 1’s lag OLS estimate different quantities, and that distinction is what determines how to read the null. The OLS compares the marginal coefficients on bearish and bullish sentiment, so the sensitivity of volume to each direction with the other held fixed (Section 4.2.3). The test here compares the mean predicted volume of bearish hours against bullish hours, which is an unconditional difference of group means with nothing held fixed at all. A larger marginal sensitivity to bearish news, which is what Phase 1 found, can sit perfectly well alongside near-equal average volume across bearish and bullish hours, for instance if bullish hours happen to cluster in higher-baseline-volume periods like risk-on sessions. So the absence of a group-mean asymmetry here does not contradict Phase 1. It answers a coarser, confounded question.

The point-prediction differences are small and non-significant at every horizon, with no

consistent sign across slices. This is not a power problem: the bearish and bullish groups each hold hundreds of samples, and no test becomes significant under alternative sentiment thresholds swept from 0.0 to 0.5 once we correct for multiple comparisons. What it actually mirrors is the target. In the real test data, the future volume of bearish and bullish hours is statistically indistinguishable at the 1h to 6h horizons, and the only significant ground-truth difference, at +12h, has bullish above bearish, a sign the model reproduces with a predicted difference of -0.244 . So the model is tracking a group-level near-symmetry that is genuinely there in its target period, not washing out a signal that exists. The one directional structure the TFT does carry is temporal rather than in magnitude: the attention pattern (Section 4.3.7.5) goes to $-6h$ for bearish sentiment and $-1h$ for bullish, matching Phase 1's +6h peak.

Synthesis on RQ2. Phase 1's lag OLS stays the primary and appropriate evidence for RQ2, because it measures the directional sensitivity the research question is actually about. TFT v2 shows no magnitude asymmetry in its point predictions, and that is the expected outcome rather than a failed replication: the group-mean comparison is a different and confounded estimand, and the model's target is itself near-symmetric across sentiment direction over the test period. What the model genuinely adds is temporal, the attention asymmetry of Section 4.3.7.5, where bearish is integrated over a longer window ($-6h$) than bullish ($-1h$). That corroborates the +6h lag structure of RQ1 without leaning on the magnitude test at all.

4.3.7.7 Success criteria evaluation

We now check TFT v2 against the three success criteria declared in Section 4.3.7.1, using the evidence from Sections 4.3.7.3 through 4.3.7.6.

TFT v2 meets the first two criteria and meets the third only in part. The Phase 2 features do drive prediction, with the channels in the top four and the entity block carrying half of total importance; the model substantially beats persistence on the aggregate liquidity targets; and its attention pattern independently corroborates Phase 1's +6h lag structure. The qualification on the third criterion is worth stating plainly rather than rounding up to a pass. As declared in Section 4.3.7.1, that criterion asked for the largest persistence reduction to fall in the +6h range, and it does not: the curve is largest at +12h and grows monotonically because persistence degrades with horizon. The underlying goal of the criterion, independent corroboration of the +6h response, is met, but by the attention diagnostic rather than by the error curve the criterion named. None of that makes Phase 2 the source of evidence for the research questions. Phase 1's lag OLS is still the primary statistical evidence for RQ1 (the +6h peak) and RQ2 (the bearish > bullish sensitivity), while TFT v2 contributes evidence that is complementary and methodologically distinct: interpretable feature importance, and a bearish-sentiment attention peak at $-6h$ confirming

Table 4.17: TFT v2 success-criteria evaluation.

Criterion	Status	Evidence
1. Channels and entities as visible drivers	Pass	<code>supply_impact</code> and <code>demand_impact</code> appear in the top four features (12.1% and 5.5%); the 71 entity flags carry 52% of total VSN importance between them, with six in the top ten; attention layer non-degenerate with peak at $-1h$; bearish/bullish attention divergence ($-6h$ vs $-1h$) matches Phase 1’s identified lag
2. Substantial persistence-relative improvement	Pass	<code>log_volume</code> MAE reduction of 46–71% across the 1–12h window and <code>amihud</code> 43–45% (<code>price_range</code> is the exception on the war regime, Section 4.3.7.3)
3. Lag structure consistent with Phase 1	Partial	Bearish-sentiment attention peaks at $-6h$, matching the lag OLS peak under a method that shares no assumptions with it. The persistence-reduction curve, however, is largest at $+12h$ (71%, against 67% at $+6h$) and rises mainly because the baseline decays, so it is consistent with a response somewhere in the 1–12h window but does not locate it at $+6h$ as the criterion specified (Section 4.3.7.5)

the same lag structure. The two phases are complementary, not in competition. The configuration evaluated here is what the rest of the thesis means by TFT v2.

4.3.8 Methodological comparison: Phase 1 versus Phase 2

The two phases come at the same research questions through structurally different methods. Phase 1 (Section 4.2) uses regression, contemporaneous and lag OLS, on FinBERT sentiment over a regex-filtered corpus. Phase 2 (Section 4.3) uses a deep-learning forecaster, TFT v2, on LLM-derived channel-decomposed features over an LLM-filtered corpus. Neither replaces the other. They answer complementary questions about the same data.

4.3.8.1 What Phase 2 changed methodologically

Phase 2 changed the analytical stack at five levels (corpus, feature extraction, tone versus price impact, entity normalization, and analytical framework), and each change answers a specific Phase 1 limitation.

Corpus expansion. Phase 1’s 13,690-article regex-filtered corpus (`body_valid`) only covers the pre-war period to February 2026. Phase 2’s LLM-filtered corpus (`usable_strict=1`, 10,514 articles) runs through May 2026, so it includes the 28 February 2026 war onset and what followed, and the `usable_strict` filter (Section 4.3.3) drops 1,161 topical-but-channel-neutral articles. The Phase 2 corpus is smaller in count but more complete in

time, because it covers a regime change Phase 1’s data simply does not contain.

Feature extraction. Phase 1’s FinBERT gives one three-class sentiment label per article (bearish, neutral, bullish), averaged hourly. Phase 2’s LLM extracts a structured schema per article: a composite sentiment score, three channels (`supply_impact`, `demand_impact`, `risk_premium`), magnitude, certainty, event type, time horizon, and entities. The channels are what let the model separate supply, demand, and risk-premium news that FinBERT flattens onto one axis.

Tone versus price impact. The two sentiment signals do not just differ in resolution, they measure different things, and the difference turns out to be diagnostic. FinBERT classifies the surface financial tone of an article. The Haiku `sentiment_score` is prompted for the net directional impact on the WTI price, with the instruction to “assess the actual content, not the headline tone”. Across the 6,300 articles scored by both methods the two agree only moderately (Pearson 0.54), and the disagreement is concentrated in one place: geopolitical news, where mean FinBERT tone is negative (-0.116) while mean Haiku price impact is bullish ($+0.137$). On demand, macro, and supply articles the two agree in sign. The mechanism is visible in the headlines they split on, for example “Russian Sanctions Drive Oil Prices Higher” (FinBERT -0.95 , Haiku $+0.85$) and “Oil prices soar after Israel launches strikes on Iran capital” (FinBERT -0.96 , Haiku $+0.80$). FinBERT anchors on the alarming vocabulary, sanctions and strikes, and labels the article negative even when the headline says outright that prices rose, because it has no asset grounding and no way to reason from a supply threat to a higher price. The LLM, prompted to think about price impact for WTI specifically, reads the same articles as bullish. This is the FinBERT limitation that motivated the Phase 2 extraction, in its most concrete form, and it is confined to exactly the geopolitical news carrying the highest risk premium and the highest trading volume (Section 4.3.7.4). It also explains how a sentiment-sign asymmetry test can come out differently across the two phases (Section 4.3.7.6): the high-volume geopolitical news lands in opposite sentiment buckets depending on which definition you use.

Entity normalization. Phase 1 has no entity features at all. Phase 2 maps entity mentions onto 71 canonical flags (Section 4.3.2 and Appendix D), giving each hour a binary vector of which actors (US, Iran, Saudi Arabia, OPEC+, and so on) the dominant article named. No Phase 1 method had access to that signal.

Analytical framework. Phase 1’s regression isolates the effect of specific variables at specific lags but produces no forecast. Phase 2’s TFT predicts three targets at four horizons jointly from the previous 48 hours, but cannot isolate individual variable effects the way OLS does. This is not a move from a weaker method to a stronger one. It is a move from one that isolates effects to one that integrates them, and each one covers the

other’s blind spot.

4.3.8.2 Why cross-phase numbers are not comparable

TFT v1 and TFT v2 loss values cannot be compared directly, because nearly everything underneath them is different: the loss function (v1 `QuantileLoss` on one target, v2 `MultiLoss` on three, so roughly three times the per-target scale), the validation period (v1 roughly 2,160 pre-war hours from an 80/20 split, v2 roughly 2,216 transitional hours from the 60/20/20 split), the horizons (v1 one hour, v2 four jointly), the feature count (v1 has 18 inputs, 5 known and 13 unknown reals, against v2’s 90), and the upstream corpus and extraction (FinBERT on the regex corpus versus LLM channels on the `usable_strict` corpus). Putting v1’s `val_loss` of 0.204 next to v2’s 0.427 would only mislead rather than inform.

The fair comparison is persistence-relative reduction on a shared target and horizon (`log_volume` at 1h), where both models beat persistence substantially and neither fails. Phase 2’s contribution is not better forecasts, it is a richer analytical vocabulary: where v1 dumps its importance onto one scalar sentiment score, v2 spreads it across decomposed channels, entity flags, and macro covariates. So the progression from Phase 1 to Phase 2 is real but qualitative and analytical, not a numerical win. The thesis’ primary contribution is the Phase 2 methodology, not a claim that it forecasts better.

4.4 Summary

Chapter 4 has reported the empirical results of both phases.

Phase 1 (Section 4.2) established the initial evidence for the research questions, through regression on a pre-war corpus of 13,690 articles with FinBERT sentiment features. The headline bias experiment (Section 4.2.2) showed that title-only and title-plus-body FinBERT inputs produce substantially different sentiment classifications, which motivated the move to richer feature extraction in Phase 2. The lag OLS (Section 4.2.3), including the contemporaneous $k = 0$ case, confirmed a positive news-volume association and put the peak of news impact on liquidity at the +6 hour lag, with the bearish coefficient exceeding the bullish one at every dominant lag, by about 18% at the peak. Both coefficients are individually significant at $p < 0.001$; the ordering between them is a consistent pattern across lags rather than the outcome of a formal equality test. An exploratory vector autoregression (Section 4.2.4) was abandoned once the sparsity of the hourly sentiment signal proved incompatible with its assumptions, which reinforced the move to the deep-learning models of Phase 2.

Phase 2 (Section 4.3) extended the analysis with LLM-based feature extraction on a corpus

covering both the pre-war and war regimes. The Schema v2 extraction produced channel-decomposed features (Section 4.3.2) separating supply-side, demand-side, and risk-premium news impacts, alongside 71 canonical entity flags from a normalized entity list. Comparing the regex-based Phase 1 filter against the LLM-based Phase 2 filter (Section 4.3.3) identified specific error modes in each and motivated the stricter `usable_strict=1` variant used for TFT v2 training. TFT v1 (Section 4.3.5) validated the deep-learning approach on a single target and horizon, beating persistence substantially and showing a two-cluster attention pattern (short-term at $-4h$, daily memory at $-27/-28h$). TFT v2 (Section 4.3.7) integrated the full set of Phase 2 investments: channel decomposition, entity flags, proper categorical encoding, multi-horizon prediction, and multi-target learning. The v2 canonical configuration cuts prediction error over persistence by 46 to 71 percent on `log_volume` across horizons, establishing that the Phase 2 features carry signal over the whole 1-to-12-hour window, though that curve rises largely because the persistence baseline decays and so does not by itself locate the response at any lag. The Variable Selection Network ranks VIX and `supply_impact` as the top two features and puts six entity flags in the top ten. The directional asymmetry test, sixteen bearish-versus-bullish comparisons of predicted volume, returned no significant group-level difference. That reflects a different and coarser estimand than Phase 1’s marginal-coefficient asymmetry rather than a failed replication, and the TFT’s complementary evidence for RQ2 is temporal: its attention goes to $-6h$ for bearish sentiment against $-1h$ for bullish.

The methodological comparison of the two phases (Section 4.3.8) argued that comparing v1 and v2 numbers directly is not appropriate, given the differences in validation sets, loss functions, prediction horizons, feature counts, and news corpus. Both models beat persistence substantially in their own settings, and Phase 2’s contribution is analytical richness rather than measurably better raw accuracy. That section also established that FinBERT’s tone-based sentiment and the LLM’s price-impact sentiment measure different constructs, agreeing overall but diverging in sign on the high-volume geopolitical news.

The success criteria set for TFT v2 before training were evaluated in Section 4.3.7.7: the first two pass, and the third is met in substance but not as specified, since the $+6h$ corroboration comes from the attention pattern rather than from the persistence-reduction curve the criterion named. The evaluation refines what the Phase 1 findings mean through a different methodological lens, and it exposes the `price_range` regime-extrapolation limitation that feeds into the discussion. Chapter 5 develops the implications of these findings along with the limitations and directions for future work. Chapter 6 concludes.

5. Discussion

This chapter interprets the findings of Chapter 4, puts them next to the prior work from Chapter 2, and sets out what the study cannot claim. It is organised around the two research questions, the lag structure of the news-liquidity response (RQ1) and its directional asymmetry (RQ2), followed by the methodological contributions of the two-phase design and the limitations that qualify everything above them.

5.1 Lag structure of the news-liquidity response (RQ1)

RQ1 asks at what lag news events hit WTI liquidity hardest. Two separate analyses land on the same answer.

- **Phase 1’s lag OLS** (Section 4.2.3) places the peak of the bearish news effect on `log_volume` at +6 hours, with the effect built up over the preceding hours and decayed to insignificance by +8 hours.
- **Phase 2’s TFT v2** corroborates this (Section 4.3.7.5) through its encoder attention, which, disaggregated by sentiment direction, peaks at −6 hours for bearish-sentiment hours, aligning exactly with the lag OLS peak. Its per-horizon error curve is a weaker instrument for this purpose: it confirms that the Phase 2 features carry predictive signal across the whole 1-to-12 hour window, but it rises mainly because the persistence baseline decays with horizon, so it does not independently locate the response at any lag.

One of these is a regression coefficient estimated on a single sentiment value at a fixed lag. The other is a deep-learning forecaster integrating a 48-hour window of many features at once. They have almost nothing in common, and they pick out the same lag, +6 *hours*. That agreement, on a point rather than a broad window, is the strongest evidence this thesis has for RQ1.

What it means is that news does not reprice liquidity on contact. It propagates over several hours. If the response were purely contemporaneous the peak would sit at lag zero, and if the market absorbed news fully and immediately there would be no predictable structure left six hours out to find.

What we see instead is a response that builds and then persists across the session and into the next hour. That fits gradual information diffusion and traders reacting at different times, rather than one clean repricing event. The TFT attention fills the picture in: a strong recency component at -1 hour, since the most recent news is always the most useful thing for an immediate forecast, sitting alongside a longer integration window that applies to bearish news specifically.

So the market seems to **take longer** to work through news implying a *downward* price move than news implying an *upward* one. That is a directional detail inside the temporal structure. Phase 1's average-lag estimate cannot see it, and the TFT's direction-conditioned attention can.

All of this is consistent with the premise from the literature (Section 2.2) that news sentiment moves trading activity and not only prices [Tetlock, 2007]. What is new here is resolution. Work at **daily** frequency cannot see an intraday six-hour propagation window at all, and the **hourly** design of this thesis exists to catch exactly that. The literature already establishes that news effects concentrate at high frequency, mostly within hours of release [Ederington and Lee, 1993]. Practically, the result says that an hourly monitor of WTI-relevant news carries liquidity information that is most actionable not in the same hour, but roughly six hours later, within the same trading day.

5.2 Directional asymmetry of the response (RQ2)

RQ2 asks whether bearish news produces a different liquidity response than comparable bullish news. Phase 1's lag OLS (Section 4.2.3) says yes: the coefficient on bearish sentiment beats the coefficient on bullish sentiment at the dominant lags, and that ordering is the thesis' primary evidence for RQ2. Two qualifications belong with it from the outset.

The first is what kind of statement it is. It is about marginal sensitivity, meaning how much volume moves for an increment of bearish rather than bullish sentiment, with the other held fixed. The second is how strong it is. Each coefficient is individually significant at the peak ($p < 0.001$) and bearish exceeds bullish at every dominant lag, by about 18% at +6h, but the difference between the two coefficients is not itself submitted to an equality test. The evidence is therefore a consistent ordering across lags, reinforced by an independent method in Phase 2's attention, rather than a single tested contrast. A formal test of $\beta_1 = \beta_2$ would sharpen this, and Section 5.5 returns to it.

Phase 2's TFT v2 does not reproduce the asymmetry in its point predictions. Across sixteen bearish-versus-bullish comparisons of predicted volume (Section 4.3.7.6), none is significant. That looks like a failed replication and it is not one, because the two analyses are estimating different quantities.

The lag OLS compares marginal coefficients, which are slopes with other variables held fixed. The TFT test compares the average predicted volume of bearish hours against bullish hours, which is an unconditional difference of group means with nothing held fixed. Those can both be true at once. A larger marginal sensitivity to bearish news can sit alongside near-equal average volume across bearish and bullish hours, for example if bullish hours happen to fall in periods where baseline volume is already high. The null is also not a power artifact. The two groups each hold hundreds of samples, the result survives sweeping the sentiment threshold from 0.0 to 0.5, and the target itself is near-symmetric: in the actual test data, future volume for bearish and bullish hours is statistically indistinguishable at the 1 to 6 hour horizons. The model even reproduces the one real difference, a bullish-above-bearish gap at +12 hours. It is tracking a near-symmetric ground truth rather than washing out a signal that was there.

There is a second reason the phases point different ways, and it runs deeper: they do not define sentiment the same way (Section 4.3.8). FinBERT scores the surface financial tone of an article. The Haiku `sentiment_score` is prompted for the net directional impact on the WTI price. The two agree only moderately (Pearson 0.54), and they disagree in sign on exactly one kind of news, which is geopolitical events.

A headline such as “Russian Sanctions Drive Oil Prices Higher” is negative in tone (sanctions, conflict) but bullish in price (a supply threat lifts crude), so FinBERT labels it bearish while Haiku labels it bullish.

That geopolitical news is the high-volume, high-risk-premium subset that dominates the liquidity response in the first place. Under Phase 1’s tone-based definition it lands in the bearish bucket, which reinforces the bearish-over-bullish asymmetry. Under Phase 2’s price-based definition it lands in the bullish bucket, which is why the TFT’s price-conditioned test finds no bearish-over-bullish gap and even a mild reversal. So the tension between the phases is not a contradiction. It is what happens when bearishness is measured two different ways.

For the TFT it is not a matter of negative versus positive news, but of the *risk premium* the news implies.

Looking at what the TFT actually leans on makes this clearer. Its most important features (Section 4.3.7.4) are VIX, the supply and demand channels, and the entity block, whose 71 flags carry 52% of total importance between them while the composite `sentiment_score` sits 21st. None of those is a bearish signal in the price sense. VIX is a volatility expectation, the channels are fundamental supply and demand judgments, and an entity flag records only that a story named a given actor, carrying no direction at all. What the model weights, in other words, is how much a story raises supply uncertainty and how salient its subject is, not which way it points. Which specific entities rank highest is not stable

enough to interpret individually (Section 4.3.7.4), so the argument rests on the block and on the feature types, not on a roster of actors. So the asymmetry the model learned is not bearish over bullish. It is risk-and-salience over baseline: WTI liquidity responds to how much a piece of news raises uncertainty about supply, not to whether it implies a higher or lower price.

The answer to RQ2 therefore depends on how sentiment is defined and which quantity is being estimated. There is a robust bearish-over-bullish asymmetry in the marginal sensitivity of volume to negative-tone news (Phase 1), and that stays the thesis' primary answer. It does not show up as a difference in the level of predicted volume between price-bearish and price-bullish hours (Phase 2), because the volume-driving geopolitical news is price-bullish, and because the model organises its predictions around risk and salience rather than price direction. Both readings are right inside their own framing. Stating the asymmetry properly means naming the sentiment construct and the estimand along with it.

5.3 Methodological contributions

Beyond the two research questions, the thesis makes five methodological contributions. Each one came out of a concrete problem we ran into, not from deciding in advance to contribute something.

Aligning a continuous news stream onto a discontinuous trading grid. Sub-daily work creates a problem that daily work does not have. News publishes around the clock; the market does not trade around the clock. So before any hourly analysis can begin, every article has to be placed on a grid punctuated by overnight, weekend and holiday closures. Two rules do that here (Section 3.2). *Ceiling assignment* sends an article published at 14:23 to the 15:00 bar rather than the 14:00 one, because the 14:00 bar had already been forming for twenty-three minutes when the article appeared, and letting news explain a candle that predates it would break the causal reading outright. *Forward-assignment* then carries off-hours articles to the next tradeable hour instead of discarding them, which matters because weekend OPEC communiqués and overnight geopolitical events are exactly the highest-impact items in the corpus. Both rules are conservative in the same direction, understating the immediate reaction rather than inflating it, so the +6h peak is if anything a lower bound on how quickly the market responds. Setting aside the boundary case below, 22.9% of articles are forward-assigned, with a median gap of half an hour and a maximum of 73.8 hours across a holiday weekend. That is the scale of the problem being solved rather than a marginal correction, and it is a prerequisite for intraday work that a daily design never has to confront.

One boundary defect belongs in this account. Forward-assignment has no natural answer

for articles published *before* the market grid begins, and the implementation carries all of them to the first available hour: 3,657 articles, 16% of the corpus, collapse onto 13 May 2024 12:00 with a mean assignment gap of 1,617 hours. That hour is index 1 of the grid and sits inside the training partition, so it never enters validation or test and reaches only the encoder window of the earliest training samples; its effect on the reported results is negligible. The correct handling is to drop pre-grid articles rather than forward-assign them, and any hourly corpus statistic evaluated at that timestamp should be read as an artefact rather than as data.

Channel decomposition as a way to calibrate model annotation. Splitting sentiment into orthogonal supply, demand, and risk-premium channels (Section 3.3.4), following the structural decomposition of oil-price shocks [Kilian, 2009], was a fix for a specific failure: the composite `sentiment_score` disagreed across model families (Haiku versus GPT-5.5) on high-magnitude geopolitical events (Section 4.3.4). Recalibrating on the channels lifted cross-model agreement on the composite from 0.39 to 0.88, with channel correlations of 0.94, 0.96, and 0.82. So the channels do two jobs: they are higher-resolution features for the TFT, and with no human ground truth available, cross-model agreement on them is a defensible validation signal for the extraction [Gilardi et al., 2023, Zheng et al., 2023].

Tone versus price impact. A tone classifier and a price-reasoning LLM are not a coarse and a fine version of the same measurement. They measure different things (Section 4.3.8, Section 5.2). FinBERT reports an article’s surface financial tone. The LLM does the asset-specific causal step, from a supply threat to a higher crude price, which a tone classifier has no way to do. The geopolitical sign-flip is the demonstration. The lesson travels beyond this thesis: for an asset-specific task, sentiment has to be defined relative to the asset and its causal mechanism rather than as generic positive-or-negative tone, and a study that mixes the two can reach opposite conclusions depending on which one it happened to pick.

LLM-as-filter for noisy news corpora. Phase 1 decided what to include with a brittle regex heuristic over the article body (Section 3.4.1). Phase 2 replaced it with the LLM’s own judgment of material relevance (`usable`). The regex is lexical and the LLM filter is semantic, and comparing them (Section 4.3.3) shows where each one breaks. The training filter, `usable_strict`, then adds a non-zero channel score on top, but it is worth keeping the two ideas apart: `usable` is the relevance judgment and the actual contribution here, while `usable_strict` is a signal requirement that drops on-topic articles which simply carry no channel information (Section 4.3.3). For web-scraped news, where relevance is about meaning rather than keywords, a semantic filter is a practical alternative to keyword heuristics and transfers to other corpora.

Phase-based progression as a reproducible template. The two-phase design is a contribution in itself. A simple interpretable baseline (FinBERT features with OLS) answers the research questions with classical statistics whose assumptions you can see. A richer extraction with a more expressive model (LLM channels with a TFT) then corroborates the baseline and adds vocabulary to the analysis. Neither phase gets discarded. The second does not supersede the first, it triangulates it, and both have their own limitations. This is a reusable template for empirical news-impact studies that want interpretability and expressiveness without giving up either one.

5.4 Limitations

Several limitations qualify the conclusions and mark out how far they reach.

Inter-model calibration is weaker than human ground truth. We validate the LLM features by cross-model agreement (Haiku versus GPT-5.5, Section 4.3.4) rather than against expert human annotation, which the project could not afford (Section 3.7). Cross-family agreement tells us something real, since two models built by different developers converging on a reading is not nothing, but it is not a gold standard. Both models can share a blind spot, and agreeing is not the same as being right. The calibration sample is also small ($n = 30$) by design, which is enough to surface systematic disagreements and not enough to estimate population agreement precisely.

Regime extrapolation on price_range. The most concrete modeling limitation is TFT v2 failing on `price_range` in the war regime (Section 4.3.7.3). Trained only on the pre-war period, the model falls back toward the historical mean when asked about a high-volatility regime it never saw, and it loses to a naive persistence baseline. This is a clean illustration of a model constrained by what its training window contained. It is also a warning not to assume the aggregate liquidity results (`log_volume`, `amihud`), which do transfer across the regime boundary, will generalize to every target.

Construct dependence of the sentiment findings. As Section 5.2 spells out, the directional conclusions depend on how sentiment is defined (tone versus price impact) and which quantity is estimated (marginal coefficient versus group mean). Stated carefully this is a strength, but it is a limitation for anyone who wants a single unconditional answer. There is no framing-free version of the asymmetry, and any restatement has to carry its definitions with it.

Corpus composition. The news corpus comes from GDELT (Section 3.1.4), a broad machine-monitored aggregator that skews toward retail-facing web and print coverage rather than the institutional feeds, trading terminals and wire services, that professional oil traders actually read. So what the models extract is the sentiment of publicly available

news, which may lead, lag, be biased, or just differ from the information set that moves institutional order flow.

The Phase 1 lag regressions cannot be re-fit from the archived data. Hourly price history is served by `yfinance` on a rolling window of roughly two years, so the market grid is not a fixed asset. Phase 1’s grid began on 11 March 2024, and by the time the data were re-pulled for Phase 2 the earliest hours had aged out; the retained hourly file now starts on 13 May 2024 (Section 4.2.1). Around 1,700 Phase 1 articles fall before that boundary, so the lag regressions of Section 4.2.3 cannot be reproduced from the repository as it stands, and the archived coefficient table is the surviving record of them. The Phase 2 results are unaffected, since they run on the retained grid. Anyone repeating this design should snapshot the price grid alongside the derived features rather than assume it can be re-fetched.

Non-stationarity and single-commodity scope. The dataset covers roughly 24 months and contains a structural break, the Iran war onset (Section 4.3.1), which makes the series non-stationary across the train/test boundary. The war-regime results come from one episode and should be read as suggestive rather than established. The study also covers one instrument, WTI crude oil futures. The channel decomposition and the risk-and-salience reading of the liquidity response are plausibly specific to a commodity with WTI’s geopolitical supply structure, and neither the lag nor the asymmetry finding can be assumed to carry to other commodities or asset classes without someone replicating it.

5.5 Future work

The limitations point to several extensions, and most of them are small enough to be tractable.

A formal test of the directional asymmetry. RQ2’s primary evidence is an ordering of two coefficients that are each significant, not a tested difference between them (Section 5.2). The direct remedy is cheap: a Wald test of $\beta_1 = \beta_2$ on the +6h regression, which the fitted model already supports since it needs only the estimated covariance of the two coefficients, reported alongside a confidence interval for $\beta_1 - \beta_2$. Bootstrapping the difference across lags would additionally show whether the ordering is stable or driven by the peak alone. This would convert the asymmetry from a consistent pattern into a quantified one, and it should be done before the result is restated anywhere as a significant difference.

Small-scale human calibration. The clearest methodological gap is the missing human ground truth (Section 5.4). The tractable next step is not annotating everything but building a small expert-annotated anchor set, say 200 articles scored by an oil-market

professional, to benchmark both the LLM extraction and the cross-model agreement against. That would turn the inter-model calibration from a standalone signal into a calibrated one, without paying to annotate the full corpus.

Separating tone and price sentiment as parallel features. The construct-dependence finding (Section 5.2) suggests an experiment, not just a caveat: carry the FinBERT tone score and the LLM price-impact score as separate features and let the model weigh them itself. For a liquidity target especially, where salience and risk drive volume more than price direction does, the tone signal may hold information the price signal does not. Their disagreement, the geopolitical sign-flip, might even work as a feature in its own right, marking high-uncertainty news.

Multi-regime training. The `price_range` extrapolation failure (Section 5.4) is a training-window problem rather than a model-capacity one. As more regime episodes accumulate in the data, retraining across multiple structural breaks, or conditioning on a regime indicator explicitly, would test whether the volatility target becomes learnable once the model has seen more than one regime.

Cross-commodity replication. The risk-and-salience reading (Section 5.2) is a hypothesis about oil specifically, where geopolitical supply risk is price-bullish. Running the two-phase design on commodities with different supply and demand structures (natural gas, gold, copper) would test whether the same risk-premium mechanism drives their news-liquidity response or whether it belongs to WTI's geopolitical exposure. Cross-commodity spillovers, where oil news moves gas or gold liquidity, are a bigger question this thesis leaves open.

Richer entity representation. The entity flags are a binary vocabulary of 71 canonical actors (Section 4.3.2). Swapping them for domain-aware entity embeddings, learned representations that would put Oman and Qatar near each other as Hormuz-adjacent producers, would let the model generalize across related actors instead of treating each one as an independent indicator. That could sharpen the risk-and-salience signal the current flags only approximate.

6. Conclusion

This thesis set out to understand and model how news propagates into the liquidity of WTI crude oil futures at hourly resolution, along two axes: the temporal lag of the response (RQ1) and its directional asymmetry (RQ2). It went at the questions in two phases, a regression-based Phase 1 on FinBERT sentiment and a deep-learning Phase 2 on LLM-extracted, channel-decomposed features.

This chapter states the contributions, summarizes the findings, and closes.

6.1 Summary of contributions

The thesis makes three empirical and five methodological contributions.

Empirically:

- **An hourly lag structure for the news-liquidity response.** The response peaks at +6 hours, having built over the preceding hours and decaying to insignificance by +8. The Phase 1 lag OLS establishes it, and the Phase 2 TFT corroborates it independently through its direction-conditioned attention, which places the bearish peak at −6 hours under a method sharing no assumptions with the regression. Daily-frequency work cannot see this structure at all.
- **A precise account of the directional asymmetry.** Bearish sentiment carries a larger marginal coefficient than bullish at every dominant lag, by about 18% at the peak (Phase 1). The ordering is consistent across lags, though the difference between the two coefficients is not itself submitted to an equality test (Section 5.5). It does not appear at all as a difference in the level of predicted volume (Phase 2), because the model organizes its predictions around risk and salience rather than bullish or bearish price direction. The precision here is as much the contribution as the result: stating the asymmetry correctly means naming the sentiment construct and the estimand it refers to.
- **A news-representation finding.** Title-only and title-plus-body FinBERT sentiment disagree on 41.6% of articles, and titles lean more bullish than the bodies they head. Which news representation you pick is itself a modeling decision.

Methodologically (developed in Section 5.3):

- **Temporal alignment of continuous news onto a discontinuous trading grid**, the prerequisite for any intraday design: a causally conservative ceiling rule plus forward-assignment of the 22.9% of articles that publish while the market is shut.
- **Channel decomposition** as a higher-resolution feature set and, through cross-model agreement, a moderately defensible validation substrate where no human ground truth exists.
- **The tone-versus-price distinction**, showing that a sentiment measure is defined by its construct, and that a tone classifier and a price-reasoning LLM part ways on exactly the geopolitical news that drives volume.
- **LLM-as-filter** as a semantically grounded alternative to regex heuristics for noisy web-scraped news corpora.
- **A two-phase template** pairing an interpretable statistical baseline with an expressive deep-learning model that corroborates it rather than replacing it.

6.2 Summary of findings

On RQ1, news hits WTI liquidity hardest not in the same hour but roughly six hours later. Two independent methods agree on this: the Phase 1 lag OLS peaks at +6 hours, and the Phase 2 TFT, trained and evaluated separately, attends to −6 hours for bearish news. Its persistence-relative gain rises across the whole 1-to-12 hour window, confirming that the Phase 2 features carry signal throughout it, but that curve is driven by the decay of the baseline and does not itself single out a lag. The liquidity response to news is gradual rather than instantaneous.

On RQ2, the answer depends on how the question is asked. Bearish sentiment carries a consistently larger marginal coefficient than bullish across the dominant lags, an asymmetry in the sensitivity of volume to negative-tone news that is the thesis’ primary answer, and one stated as a consistent ordering rather than a tested contrast. It does not survive as a difference in the average level of predicted volume, for two reasons: the model organizes its forecasts around the risk and salience a news item carries rather than its price direction, and a tone-based and a price-based sentiment measure disagree on exactly the geopolitical news that moves the market. WTI liquidity responds to how much a piece of news raises supply uncertainty more than to whether it reads as nominally good or bad.

As a forecasting exercise, TFT v2 cut `log_volume` error over a persistence baseline by 46 to 71 percent across horizons and `amihud` by 43 to 45 percent, with VIX, the supply and demand channels, and the entity flags as a block, which carry 52 percent of total feature importance between them, as what it leans on. It failed only on `price_range` in the unseen

war regime, which is a clean case of regime extrapolation. Alongside the two headline answers, three supporting findings shaped how the research developed: a 41.6 percent title-versus-body sentiment disagreement, a composite-sentiment calibration failure between model families that motivated the channel decomposition, and a semantic-versus-lexical filter comparison that motivated the `usable_strict` rule.

6.3 Closing remarks

This thesis started from a simple observation, that news moves markets, and asked something more specific of it: at what lag, and with what asymmetry, does news move the liquidity of WTI crude oil at hourly resolution. The answer has two parts. Empirically, the response is gradual and peaks about six hours after publication, and its directional asymmetry is consistent but has to be stated carefully, because it lives in the sensitivity of volume to news rather than in the average volume of nominally good or bad hours. Methodologically, getting to that precision meant treating sentiment not as a single quantity handed to us but as a construct to define, decompose, and validate. That is what the channel decomposition, the LLM extraction, and the two-phase design are for.

If one idea carries beyond this study, it is that for a commodity like crude oil the liquidity response to news is organized around risk rather than direction. The market does not trade more because the news is bad. It trades more because the news raises uncertainty about supply, and in the oil market that uncertainty is often, against intuition, bullish for price. Once that is clear, the apparent tension between a tone-based and a price-based reading of the same events goes away, and it changes what a news-driven liquidity model should be built to detect.

The work is bounded: one commodity, two years, one structural break, and a validation resting on model agreement rather than human judgment. Within those bounds it offers a concrete empirical result and a reusable way of getting there, and the extensions in Section 5.5 are mostly small steps rather than new programs. How news propagates into market liquidity is a question of resolution as much as of theory, and at hourly resolution it shows more structure, and more nuance, than a daily view can.

A. Extraction prompt

This appendix reproduces the Haiku v2 extraction interface: the system prompt, the `extract_article_features` tool schema passed to the tool-use API, and the request configuration. The conceptual field-level schema and its revision history are in Appendix B. The source of record is `03_src/nlp/llm_features.py`. Dashes in the reproduced prompt are normalised to ASCII for typesetting.

A.1 System prompt. The following text is passed as the cached system prompt.

```
You are a financial analyst specializing in WTI crude oil markets
. Your task is to analyze news articles and extract structured
features for use in a quantitative liquidity impact model.

First, determine whether the article contains substantive WTI-
relevant content (usable=true). Set usable=false -- and return
only that field -- for: paywall placeholders, Cloudflare blocks
, cookie notices, or articles that are clearly not about oil
markets, macro, or geopolitics relevant to oil.

When usable=true, extract the remaining fields:
- sentiment_score: NET directional impact on WTI price. -1.0 =
  strongly bearish, +1.0 = strongly bullish. Assess the actual
  content, not the headline tone.
- supply_impact: Direction of impact on WTI supply. Score the
  article's implication for the quantity of oil available to the
  market.
+1.0 = strong supply increase (OPEC adds barrels, sanctions
  lifted on a major producer, new pipeline online, Venezuela
  ramping up).
+0.5 = moderate supply increase (single-country production
  gains, inventory builds).
0.0 = no supply implication.
-0.5 = moderate supply decrease (refinery outages, minor OPEC
  cuts).
-1.0 = strong supply decrease (major OPEC cuts, infrastructure
  attacks, embargoes, refinery fires affecting multiple
  facilities).
```

- demand_impact: Direction of impact on WTI demand. Score the article's implication for oil consumption.
 - +1.0 = strong demand increase (China stimulus, Fed pivot to easing, growth surprise, summer driving acceleration).
 - +0.5 = moderate demand increase (regional growth indicators).
 - 0.0 = no demand implication.
 - 0.5 = moderate demand decrease (mild slowdown signals).
 - 1.0 = strong demand decrease (recession confirmed, Fed strongly hawkish, major economy lockdowns, demand destruction events).
- risk_premium: Change in geopolitical or operational risk premium for oil. Score the direction of risk change, not the absolute level.
 - +1.0 = major escalation (military strike on oil infrastructure, war, Strait of Hormuz incident, major sanctions imposed).
 - +0.5 = moderate escalation (rising tensions, threats, troop movements).
 - 0.0 = no change in risk environment.
 - 0.5 = moderate de-escalation (talks resuming, partial sanctions lifted).
 - 1.0 = major de-escalation (ceasefire signed, diplomatic breakthrough, full sanctions removal).
- magnitude: how market-moving is this event. 0.0 = negligible, 1.0 = historic. Most articles score 0.1-0.4. Major OPEC cut = 0.9. Geopolitical crisis = 1.0.
- event_type: 1-3 categories ordered by salience from: geopolitical, supply, demand, macro, technical, other. EIA inventory articles classify as supply or macro -- do not use "inventory".
- entities: specific named actors central to the article (not incidentally mentioned). Include countries, organizations, oil companies, and key officials. Use the names exactly as they appear in the article.
- certainty: how confirmed is the information. 0.0 = rumor, 0.5 = analyst forecast, 0.9 = official announcement.
- time_horizon: immediate (hours to 1 day), short_term (days to weeks), or structural (months-plus or permanent themes such as energy transition).

The three impact fields (supply_impact, demand_impact, risk_premium) are intentionally orthogonal. An article can

affect multiple channels independently. Score each based on what the article actually says about that specific channel. If the article has no clear implication for a channel, score it 0.0.

A.2 Tool schema. The model is forced to call a single tool, `extract_article_features`, whose `input_schema` is reproduced below. Only `usable` is required; every other field is omitted when `usable=false`.

```
{
  "name": "extract_article_features",
  "description": "Extract structured WTI-relevant features from a
                  news article
                  for use in a quantitative liquidity impact
                  model.",
  "input_schema": {
    "type": "object",
    "properties": {
      "usable": { "type": "boolean" },
      "sentiment_score": { "type": "number" },
      "supply_impact": { "type": "number", "minimum": -1.0, "
                        maximum": 1.0 },
      "demand_impact": { "type": "number", "minimum": -1.0, "
                        maximum": 1.0 },
      "risk_premium": { "type": "number", "minimum": -1.0, "
                        maximum": 1.0 },
      "magnitude": { "type": "number" },
      "event_type": { "type": "array",
                      "items": { "type": "string",
                                "enum": ["geopolitical", "supply", "
                                         demand",
                                         "macro", "technical", "other"]
                                },
                      "minItems": 1, "maxItems": 3 },
      "entities": { "type": "array", "items": { "type": "
                    string" } },
      "certainty": { "type": "number" },
      "time_horizon": { "type": "string",
                        "enum": ["immediate", "short_term", "
                                structural" ]
                        },
    },
    "required": ["usable"]
  }
}
```

```
}  
}
```

Each numeric field carries a per-field **description** in the live schema that restates the scoring anchors of the system prompt; those are elided here for brevity and are recoverable from the source file. Note that `time_horizon` is a single categorical label, not an array (only `event_type` is multi-valued).

A.3 Request configuration. Requests are submitted through the Anthropic Batches API, one request per article, with the following parameters:

- `model`: `claude-haiku-4-5`
- `max_tokens`: 500
- `system`: the A.1 prompt, sent with `cache_control` ephemeral so the prompt is cached across the batch
- `tools`: the A.2 tool; `tool_choice` set to force `extract_article_features`
- user message: the title and body, where the body is UTF-8 sanitised and truncated to the first 1,500 characters (title only when no body was retrieved)

B. LLM extraction schema (v1 vs v2)

The Haiku extraction schema was revised once during Phase 2, from Schema v1 (used for TFT v1, Section 4.3.5) to Schema v2 (the canonical extraction used for all reported Phase 2 results, Section 4.3.6). The revision was the direct response to the inter-model calibration finding of Section 4.3.4: the composite `sentiment_score` was unreliable on high-magnitude geopolitical events because it conflated event valence with directional price impact.

Field-by-field comparison.

Table B.1: Schema v1 versus v2, field by field.

Field	Schema v1	Schema v2
<code>sentiment_score</code>	continuous $[-1, +1]$	continuous $[-1, +1]$ (retained)
<code>supply_impact</code>	n/a	continuous $[-1, +1]$ (new)
<code>demand_impact</code>	n/a	continuous $[-1, +1]$ (new)
<code>risk_premium</code>	n/a	continuous $[-1, +1]$ (new)
<code>magnitude</code>	$[0, 1]$	$[0, 1]$
<code>certainty</code>	$[0, 1]$	$[0, 1]$
<code>event_type</code>	single categorical label	array of one to three labels
<code>time_horizon</code>	categorical label	categorical label
<code>price_direction</code>	categorical label	dropped
<code>entities</code>	list of canonical actors	list of canonical actors
<code>usable</code>	implicit (prompt only)	required boolean field

Changes and rationale.

1. **Three orthogonal economic channels added** (`supply_impact`, `demand_impact`, `risk_premium`, each on $[-1, +1]$). This decomposes the single sentiment judgment into its separable economic drivers, mirroring the standard structural decomposition of oil-price shocks into supply, aggregate-demand, and precautionary (risk) components [Kilian, 2009]. Re-calibration lifted the composite `sentiment_score` cross-model correlation from 0.39 to 0.88 and gave the channels correlations of 0.94 (supply), 0.96 (demand), and 0.82 (risk) (Section 4.3.6).
2. **usable promoted to a required field** (from an implicit prompt instruction), enabling the LLM-based filter and the derived `usable_strict` variant (Section 4.3.3).

3. **event_type** changed from a single label to an array of one to three salience-ordered labels, so an article that is jointly geopolitical and supply-related (e.g. Iran sanctions) need not be forced into a single category.
4. **price_direction** dropped: it agreed with the sign of **sentiment_score** on all but roughly 16 of 12,024 articles and carried about 0.008 feature importance in TFT v1, and the three channels provide a richer directional substrate.
5. **sentiment_score** retained (not replaced by the channels), for continuity with the headline-bias experiment (Section 4.2.2) and as a hedge against individually noisy channels.

TFT v1 (Section 4.3.5) was trained on Schema v1; TFT v2 (Section 4.3.7) and all reported Phase 2 results use Schema v2.

C. TFT v2 ablation and hyperparameters

The reported TFT v2 (variant v2.2) was selected through a three-variant ablation, all on the 60/20/20 split with `usable_strict=1`, dropout 0.15, and early-stopping patience 10; only the design axis changes between rows.

Table C.1: Three-variant TFT v2 ablation.

Variant	Categorical encoding	Targets $\times$ horizons	Entity flags	Val loss	MAE@1h	Feature #
v2.0	int-encoded	<code>log_volume</code> $\times$ 1h	no	0.257	0.455	sentiment
v2.1	proper categoricals	<code>log_volume</code> $\times$ 1h	no	0.258	0.424	demand_i
v2.2 (canonical)	proper categoricals	$3 \times [1, 3, 6, 12]$	71	0.427*	0.585	vix 0.188

* v2.2's `val_loss` (`MultiLoss` over three targets) is not comparable to the single-target v2.0/v2.1. The key finding is that proper categorical encoding (v2.0 to v2.1) is what unlocks the channels' predictive role: `demand_impact` jumps to 0.485. The full per-run record is in `05_reports/v2_training_results.md`.

Canonical TFT v2 hyperparameters. Hidden size 32, attention head size 4, hidden-continuous size 16, dropout 0.15, learning rate 1e-3 (Adam, on-plateau patience 3), gradient clip 0.1, early-stopping patience 10, seed 42; 298,329 trainable parameters; encoder length 48h; targets [`log_volume`, `amihud`, `price_range`]; horizons [1, 3, 6, 12]; split 60/20/20 (TRAIN_END 6739, VAL 6787–9014, TEST 9062–11232).

D. Canonical entity list

The 71 canonical entities used as TFT v2 entity flags (from `03_src/nlp/llm_features.py`), each mapped from raw article mentions via the normalization of Section 4.3.2 (alias mappings omitted here for brevity):

US, Iran, Russia, China, Israel, India, Saudi Arabia, Ukraine, Venezuela, Canada, Iraq, Nigeria, UAE, Kazakhstan, Qatar, Oman, Japan, Kuwait, Pakistan, Libya, Mexico, Azerbaijan, Yemen, Lebanon, Brazil, South Korea, Guyana, UK, Algeria, Germany, Australia, Hungary, Egypt, Türkiye, Strait of Hormuz, Middle East, Gaza, Red Sea, Persian Gulf, Gulf of Mexico, Permian Basin, Europe, Asia, Trump, Maduro, Putin, Powell, Bessent, Biden, OPEC+, OPEC, Fed, EU, EIA, IEA, API, ECB, Hamas, Hezbollah, Houthis, Saudi Aramco, Chevron, Shell, BP, ExxonMobil, Rosneft, Lukoil, TotalEnergies, WTI, Brent, S&P 500.

E. Inter-model calibration comparison

This appendix gives the article-by-article record behind the inter-model calibration of Sections 3.7 and 4.3.4. The same 30-article stratified sample was scored independently by Claude Haiku 4.5 (the extraction model, denoted H) and by GPT-5.5 (an independent reference from a different model family, accessed through the ChatGPT interface, denoted G), using the identical Schema v2 prompt and tool of Appendix A. Two independently trained models agreeing is a defensible, if weaker than human, validation signal; disagreement flags genuine ambiguity in the article.

Aggregate agreement (Schema v2). The two models agree on the binary `usable` flag for 27 of 30 articles; the three disagreements (IDs 13331, 10308, 10211) are borderline single-company or forecast pieces that Haiku filtered out and GPT retained. On the 14 articles both models judged usable, the composite `sentiment_score` reaches a Pearson correlation of 0.88, with a single sign disagreement (1/14, using a 0.1 neutral band, so only scores with magnitude above 0.1 count as directional). The three channels reach Pearson correlations of 0.94 (supply), 0.96 (demand), and 0.82 (risk). For contrast, Schema v1, which scored only the composite, gave a sentiment correlation of 0.39 with 4 sign disagreements out of 13; the channel decomposition of Schema v2 is what lifted agreement (Section 4.3.6).

Usability and composite sentiment (all 30 articles).

Table E.1: Per-article usability and composite sentiment, Haiku (H) versus GPT (G).

ID	Title	Us. H	Us. G	Sent. H	Sent. G
11630	Top Stock Movers Now: Vertex, DoorDash, Con...	×	×		
16558	Oil prices fall as analysts forecast an increa...	✓	✓	-0.60	-0.68
18615	WTI Forecast: Retreats from 7-month top, h...	✓	✓	+0.70	+0.74
13331	Crude Oil Weekly Forecast 10/08: Tempting S...	×	✓		-0.42
13772	Focus on Nigeria Non-oil Revenue Potential	×	×		

Table E.1 (continued)

ID	Title	Us. H	Us. G	Sent. H	Sent. G
4108	Oil up on Opec+ meeting, summer driving seaso...	✓	✓	+0.60	+0.81
8677	Top 5 things to watch in markets in the week a...	✓	✓	+0.00	-0.16
3151	Oil recovers after surprise drawdown in US sto...	✓	✓	+0.30	+0.37
3779	Global supply chains now in Goldilocks Zone...	✓	✓	+0.40	+0.21
6068	Oil climbs with U.S. Federal Reserve pivot	✓	✓	+0.60	+0.58
3335	Gran Tierra Energy Inc. Announces First Quart...	✓	✓	+0.30	-0.06
10308	Vermilion Energy Inc. Announces Results for t...	×	✓		-0.02
7834	Is Occidental Petroleum Corporation (OXY) th...	✓	✓	+0.20	+0.17
7188	ALPS Advisors Inc. Expands Stake in En- Link Mi...	×	×		
10989	Why Occidental Petroleum Stock Plunged 10% To...	✓	✓	-0.30	+0.09
10799	Is Atmos Energy Corp. (ATO) The Best Defens...	×	×		
13211	Baytex Energy Corp.: Baytex Delivers Solid Se...	✓	✓	+0.30	-0.12
20149	Why Halliburton (HAL) Stock Is Trading Up To...	✓	✓	+0.70	+0.86
10211	Is Chevron Corporation (CVX) The Best Crude...	×	✓		-0.35
6161	1 Growth Stock Down 18% to Buy Right Now	×	×		
8882	Petroleum prices increase ahead of Christ- mas	×	×		
6160	Stock market today: US stocks slip as traders...	×	×		
13217	The Latest: Market slumps, Gaza blood- bath an...	×	×		
11929	Oil Prices Rise Amid US-China Trade Opti- mism...	×	×		
16355	Iran Warns US Intervention Would Destabi- lize R...	✓	✓	+0.30	+0.25
21358	How China built an energy fortress to survive o...	×	×		
21692	Global oil prices surge above \$111 per bar- rel...	✓	✓	+0.70	+0.85

Table E.1 (continued)

ID	Title	Us. H	Us. G	Sent. H	Sent. G
1636	Gasoline Market 2024-2032: Industry Re- port...	×	×		
19074	Governments Move to Shield Economies as Oil Ju...	×	×		
18	Oil jumps 1% in New Year after US forces repe...	×	×		

Channel scores (14 articles both models judged usable).

Table E.2: Per-article channel scores on the both-usable articles, H versus G.

ID	Supply H	Supply G	Demand H	Demand G	Risk H	Risk G
16558	+0.70	+0.72	-0.40	-0.34	-0.30	+0.18
18615	+0.00	-0.12	+0.00	+0.00	+0.60	+0.78
4108	-0.50	-0.72	+0.50	+0.48	+0.00	+0.00
8677	+0.00	+0.00	+0.00	-0.22	+0.00	+0.08
3151	-0.50	-0.44	+0.00	-0.12	+0.00	+0.00
3779	+0.00	+0.00	+0.50	+0.34	+0.00	+0.00
6068	+0.00	-0.08	+0.70	+0.52	+0.00	+0.00
3335	+0.40	+0.29	+0.00	+0.00	+0.00	+0.00
7834	+0.00	-0.12	+0.00	+0.00	+0.00	+0.44
10989	+0.00	+0.06	+0.00	+0.00	+0.00	+0.00
13211	+0.40	+0.33	+0.00	+0.00	+0.00	+0.00
20149	+0.00	-0.48	+0.00	+0.00	+0.50	+0.91
16355	+0.00	+0.00	+0.00	+0.00	+0.50	+0.45
21692	-0.60	-0.75	+0.00	+0.00	+0.40	+0.60

The residual disagreement concentrates in the risk channel ($r = 0.82$, the lowest of the three): on IDs 16558, 20149, and 7834 the two models weight the geopolitical or operational risk implication quite differently, whereas supply and demand track closely. This is consistent with risk being the hardest channel to score, noted in Section 4.3.6.

F. Extended metrics tables

This appendix reports the full TFT v2 test-set metrics summarised in Section 4.3.7.3. For each of the three targets it gives mean absolute error (MAE) and root mean squared error (RMSE) at each forecast horizon against the persistence baseline ($\hat{y}_{t+k} = y_t$, the last observed value carried forward), broken out by the full test set and by the pre-war and war regimes (split at the war onset, decoder time index 10056). “Reduction” is the percentage by which the TFT lowers MAE relative to persistence; a positive value means the model beats persistence. The test set has 2,159 forecast origins; the pre-war and war row counts differ slightly across horizons because a fixed origin’s $+k$ -step target crosses the war boundary at different k .

Table F.1: `log_volume` extended metrics.

Horizon	Slice	n	TFT MAE	Persist. MAE	Reduction	TFT RMSE	Persist. RMSE
+1h	overall	2159	0.585	1.076	+46%	0.915	2.334
+1h	pre-war	994	0.536	1.075	+50%	0.863	2.233
+1h	war	1165	0.628	1.076	+42%	0.957	2.417
+3h	overall	2159	0.577	1.452	+60%	0.895	2.734
+3h	pre-war	992	0.532	1.480	+64%	0.855	2.664
+3h	war	1167	0.614	1.428	+57%	0.928	2.793
+6h	overall	2159	0.602	1.820	+67%	0.932	2.838
+6h	pre-war	989	0.543	1.935	+72%	0.865	2.841
+6h	war	1170	0.652	1.724	+62%	0.985	2.835
+12h	overall	2159	0.631	2.174	+71%	0.958	3.062
+12h	pre-war	983	0.555	2.319	+76%	0.874	3.111
+12h	war	1176	0.695	2.053	+66%	1.024	3.021

Reading. Three patterns stand out. On `log_volume`, the primary liquidity target, the TFT beats persistence at every horizon and in both regimes, and the margin widens with horizon (from 46% at +1h to 71% at +12h overall), consistent with the delayed-propagation reading of RQ1 and stable across the war boundary. On `amihud`, the TFT also beats persistence throughout (42 to 46% overall); absolute errors are on the order of 10^{-4} and are dominated by war-regime spikes. On `price_range`, the picture splits by regime: the TFT beats persistence in the pre-war window (reduction rising to +43% at +12h) but underperforms it overall and in the war regime (negative reductions). Trained only on pre-war data, the model reverts toward the historical mean and cannot reach the war-regime volatility spikes (actual `price_range` up to 14.4 against a predicted ceiling near 1.5), the regime-extrapolation failure discussed in Section 4.3.7.3 and Section 5.4.

Table F.2: amihud extended metrics (MAE and RMSE in units of 10^{-3}).

Horizon	Slice	n	TFT MAE	Persist. MAE	Reduction	TFT RMSE	Persist. RMSE
+1h	overall	2159	0.23	0.41	+42%	2.73	3.86
+1h	pre-war	994	0.10	0.15	+31%	0.60	0.87
+1h	war	1165	0.35	0.62	+45%	3.67	5.20
+3h	overall	2159	0.23	0.41	+44%	2.72	3.86
+3h	pre-war	992	0.10	0.15	+36%	0.59	0.87
+3h	war	1167	0.34	0.62	+46%	3.65	5.19
+6h	overall	2159	0.22	0.41	+45%	2.72	3.86
+6h	pre-war	989	0.09	0.15	+37%	0.59	0.87
+6h	war	1170	0.33	0.62	+46%	3.65	5.19
+12h	overall	2159	0.22	0.41	+45%	2.72	3.86
+12h	pre-war	983	0.09	0.15	+38%	0.60	0.87
+12h	war	1176	0.33	0.62	+46%	3.64	5.17

Table F.3: price_range extended metrics.

Horizon	Slice	n	TFT MAE	Persist. MAE	Reduction	TFT RMSE	Persist. RMSE
+1h	overall	2159	0.718	0.495	-45%	1.287	0.981
+1h	pre-war	994	0.154	0.180	+14%	0.256	0.278
+1h	war	1165	1.200	0.764	-57%	1.736	1.310
+3h	overall	2159	0.719	0.578	-24%	1.286	1.107
+3h	pre-war	992	0.155	0.220	+30%	0.257	0.329
+3h	war	1167	1.199	0.883	-36%	1.733	1.475
+6h	overall	2159	0.721	0.630	-14%	1.289	1.194
+6h	pre-war	989	0.156	0.233	+33%	0.256	0.353
+6h	war	1170	1.198	0.966	-24%	1.734	1.589
+12h	overall	2159	0.724	0.701	-3%	1.294	1.252
+12h	pre-war	983	0.157	0.278	+43%	0.257	0.390
+12h	war	1176	1.198	1.055	-14%	1.737	1.658

G. Reproducibility statement

The Phase 2 training was run on Google Colab (T4 GPU); the LLM extraction and the Phase 1 analyses were run locally (Apple Silicon, MPS backend). Software with a paper of record is cited; the rest are attributed here by name, version, and source.

Software libraries. Version constraints are pinned in the training notebook `02_notebooks/13_tft_v2_training_colab.ipynb`.

Table G.1: Software libraries and versions.

Library	Version	Role
Python	3.x	runtime
numpy	< 2.0	numerics
pandas	n/a	data processing and hourly aggregation
PyTorch (torch)	$\geq 2.3, < 2.4$	deep-learning backend
Lightning (lightning)	$\geq 2.2, < 2.4$	training loop
pytorch-forecasting	$\geq 1.0, < 1.2$	TFT implementation
HuggingFace transformers	n/a	FinBERT inference [Wolf et al., 2020]
yfinance	n/a	market-data retrieval
beautifulsoup4	n/a	article-body HTML parsing
scipy	n/a	statistical tests (Welch t-test, correlations)
sqlite3 (stdlib)	n/a	database

Models and APIs.

- Claude Haiku 4.5 (Anthropic), via the tool-use API, for Phase 2 feature extraction (Section 3.3).
- GPT-5.5 (OpenAI), via the ChatGPT interface, as the inter-model calibration reference (Section 3.7).

Data sources.

- GDELT 2.0 Document API [Leetaru and Schrodt, 2013], for news article metadata (<https://www.gdeltproject.org>).
- U.S. Energy Information Administration (EIA) v2 API, weekly crude inventory series WCRSTUS1 (<https://www.eia.gov/opendata/>).
- yfinance / Yahoo Finance, hourly OHLCV for $CL=F$, $DX=Y.NYB$, and $\sim VIX$.

Determinism. The reported TFT v2 run fixes the random seed to 42 with deterministic algorithms; its exact configuration is stored in `run_metadata.json`. Exact pinned library versions and data-source access dates are to be recorded from the environment at finalisation.

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